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# FOAM: Frequency and Operator Error-Based Adaptive Damping Method for Reducing Staleness-Oriented Error for Shampoo

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## Abstract

Shampoo has attracted considerable attention for its strong empirical performance on optimization benchmarks, but its practical use is limited by the high cost of computing inverse roots of Kronecker factors. To reduce this overhead, practitioners often reuse *stale* preconditioners, creating a trade-off between computational efficiency and preconditioner fidelity. In this work, we study staleness through the lens of inverse-preconditioner error. We show that stale updates increase the discrepancy between fresh inverse-root preconditioners and their reused counterparts, and that this discrepancy propagates into stability-related quantities and regret-related bounds. This motivates using damping as a control variable to reduce inverse-root sensitivity under staleness. Guided by this analysis, we propose FOAM, an adaptive algorithm that controls both the damping level and the eigendecomposition frequency using a proxy for the staleness-oriented error. Experiments on several benchmarks show that FOAM improves the wall-clock/performance trade-off relative to Shampoo baselines while remaining robust across a range of hyperparameter settings.

## 1. Introduction

The advancement of deep learning has been propelled by massive scaling in both model architectures and datasets (Kaplan et al., 2020; Hoffmann et al., 2022; Achiam et al., 2023). From an optimization standpoint, classical first-order methods—such as SGD (Robbins & Monro, 1951) and Momentum (Polyak, 1964)—remain the default choice thanks to their low per-iteration overhead; yet, they exhibit slow convergence rates (Bottou et al., 2018). In partic-

ular, when the loss landscape is ill-conditioned (*i.e.*, exhibits highly non-uniform curvature across parameter dimensions) (Li et al., 2018; Yao et al., 2020), purely first-order updates may require a large number of iterations to make meaningful progress (Garrigos & Gower, 2023). This limitation has renewed interest in optimization approaches that incorporate richer curvature information while remaining computationally efficient, with the goal of reducing wall-clock training time.

To address these geometric challenges, preconditioned gradient methods (*e.g.*, leveraging Hessian (Wall, 1948), Fisher information (Amari, 2016), Gauss-Newton method (Buffelli et al., 2024), second-moment estimator (Duchi et al., 2011)) have emerged as a promising solution. By employing a preconditioner matrix to rescale the gradient based on local curvature, this method effectively normalizes the loss landscape, thereby maximizing progress at each update step. While widely adopted for their computational efficiency, these methods typically rely on diagonal approximations to perform element-wise scaling updates, thereby overlooking potential parameter correlations.

To overcome the limitations of diagonal approximations, structured preconditioning — such as Kronecker product-based (Martens & Grosse, 2015; Gupta et al., 2018) or layerwise adaptivity (You et al., 2020) — has been proposed to capture missing correlations efficiently. Notably, Shampoo (Gupta et al., 2018; Shi et al., 2023) captures within-layer parameter correlations via Kronecker-factored statistics, thereby achieving the superior performance demonstrated in recent benchmarks (Dahl et al., 2023; Kasimbeg et al., 2025).

However, Shampoo fundamentally requires computing the inverse  $p$ -th roots of the Kronecker factors. It is well-known that such inverse operations incur high computational costs. To address computational overhead, practitioners relied on a heuristic strategy—referred to as *stale* Shampoo—in which those expensive operations are performed at a fixed frequency. Although it is practically effective, this choice of frequency remains a heuristic without theoretical analysis.

In this work, we study staleness (*i.e.*, the delay in updating preconditioners) through the lens of inverse-preconditioner

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error. Our analysis shows that stale updates enlarge the discrepancy between the fresh inverse-root preconditioners and their reused counterparts. This discrepancy enters stability-related quantities and convergence-related bounds, motivating the use of damping as a control variable under staleness. In particular, when damping is too small relative to the measured operator gap, the inverse-root map becomes increasingly sensitive to stale drift, thereby making the resulting updates less stable. To mitigate this, we propose an adaptive algorithm that improves robust stability by dynamically modulating both the damping factor and the update frequency based on real-time error estimates.

**Contributions.** The main contributions are as follows:

- We provide a theoretical characterization of how staleness enlarges inverse-preconditioner error in Shampoo. In particular, we show how this error enters operator-gap bounds and discounted-regret analysis, and we identify damping as a control variable that reduces the sensitivity of the inverse-root map to stale drift. To the best of our knowledge, this is the first work to explicitly study damping as a mechanism for controlling staleness-oriented inverse-preconditioner error in Shampoo.
- Drawing on matrix perturbation theory, we propose FOAM, an adaptive algorithm that adjusts the damping level and the refresh decisions using an inexpensive proxy for the relative inverse-root operator error. The resulting controller balances two competing effects: suppressing staleness-related operator errors and preserving preconditioning’s effectiveness when excessive damping would otherwise weaken it.
- We evaluate FOAM on several benchmarks and show that it achieves a favorable wall-clock/performance trade-off relative to stale Shampoo and related baselines across a broad range of hyperparameter settings<sup>1</sup>. Additional analyses examine the adaptive damping dynamics, the reduction in the eigendecomposition frequency, and the contributions of the method’s main components.

## Conflict of Interest Disclosure 2. Related Work

**Preconditioned gradient methods.** Preconditioned gradient methods can be viewed as gradient descent on reparameterized variables (Grosse, 2022). Classical approaches (e.g., Newton (Nesterov et al., 2018) and natural gradient (Amari, 2016)) leverage the Hessian or Fisher information, while widely used adaptive methods such as AdaGrad/Adam (Duchi et al., 2011; Kingma, 2014) typically

<sup>1</sup>We provide our official code in <https://github.com/REAL-KENTECH/FOAM.git>

approximate (empirical) curvature using diagonal statistics (Kunstner et al., 2019). Shampoo (Gupta et al., 2018; Shi et al., 2023) advances this line by employing Kronecker-factored, block-structured second-moment matrices, capturing within-layer correlations.

**Shampoo family.** A growing family of methods builds on Shampoo’s Kronecker structure, including CASPR (Duvvuri et al., 2024), SOAP (Vyas et al., 2025), Muon (Jordan et al., 2024), SPlus (Frans et al., 2026), and KL-Shampoo (Lin et al., 2026). CASPR improves approximation quality via Kronecker-sum combinations of axis preconditioners (Duvvuri et al., 2024), but still requires expensive inverse(-root) computations. SOAP and related implementations often reuse stale eigenspaces and periodically refresh them via iterative schemes (e.g., QR) (Vyas et al., 2025); this implicitly assumes eigenspace stability, which can break when eigengaps or damping are small. Moreover, refresh triggers based on diagonalization residuals (Eschenhagen et al., 2026) do not track the inverse-root *operator* error that actually governs the update and can therefore be inefficient or miscalibrated.

**Theoretical analysis and hyperparameters.** Recent theory formalizes Shampoo-type updates through non-Euclidean steepest descent / proximal and LMO viewpoints (Bernstein & Newhouse, 2024; Pethick et al., 2025), and studies structured-preconditioner regret bounds (Xie et al., 2025). The inverse-root exponent  $p$  is another key choice as explained by the Kronecker approximation quality (Morwani et al., 2025). Damping is routinely introduced for stable inverse computations (Moré, 2006; Martens & Grosse, 2015; Ishikawa & Karakida, 2024), but has received less direct focus than step size (Anil et al., 2020; Agarwal et al., 2020; Zhang et al., 2025) or batch size (Ishikawa & Karakida, 2024; Kim & hwan Oh, 2025; Marek et al., 2026). Lastly, Damaskinos et al. (2018) dampens **stale gradients** in asynchronous SGD via a scalar weight, whereas our work controls **stale preconditioners** in Shampoo by adaptively tuning the inverse-root damping  $\epsilon$ .

## 3. Preliminary

This section formalizes the notation and stale Shampoo. For a more detailed background, we refer to Appendix B.2.

### 3.1. Notation and Shampoo Optimizer

In this section, we establish notation and introduce the Shampoo optimizer, along with its stale-update mechanism.

**Notations.** The vector space of  $m \times n$  is denoted by  $\mathbb{R}^{m \times n}$ ;  $\mathbf{0}_m \in \mathbb{R}^{m \times m}$  is the zero matrix, and the  $m \times m$  identity matrix is denoted by  $\mathbf{I}_m$ . For  $A \in \mathbb{R}^{m \times n}$ ,  $\|A\|_F$  and  $\|A\|_2$  represent its Frobenius and spectral norms, respectively.

**Algorithm 1** Shampoo with Modular Factor-wise  $\mathcal{U}(\cdot)$ 

**Require:** Learning rate  $\eta$ ; momentum coefficient  $\beta \in (0, 1)$ ; total steps  $T$ ; check period  $\mathbf{f}$ ; factor-wise update rule  $\mathcal{U}(\cdot)$ ; inverse-root power  $p$ ; base damping  $\epsilon_0$

- 1: **Initialize:**  $W_1, L_0 = \mathbf{0}_m, R_0 = \mathbf{0}_n$ , cached states  $\mathcal{S}_0^L, \mathcal{S}_0^R, \epsilon_0^L \leftarrow \epsilon_0, \epsilon_0^R \leftarrow \epsilon_0$
- 2: **for**  $t = 1$  **to**  $T$  **do**
- 3:   Receive a loss function  $f_t(\cdot) : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$
- 4:    $G_t = \nabla f_t(W_t)$
- 5:    $L_t = \beta L_{t-1} + (1 - \beta) G_t G_t^\top$
- 6:    $R_t = \beta R_{t-1} + (1 - \beta) G_t^\top G_t$
- 7:   */\* Update rule: Stale or FOAM \*/*
- 8:    $(\hat{L}_t^{-1/p}, \mathcal{S}_t^L, \epsilon_t^L) = \mathcal{U}(L_t, \mathcal{S}_{t-1}^L, \epsilon_{t-1}^L, t)$
- 9:    $(\hat{R}_t^{-1/p}, \mathcal{S}_t^R, \epsilon_t^R) = \mathcal{U}(R_t, \mathcal{S}_{t-1}^R, \epsilon_{t-1}^R, t)$
- 10:    $W_{t+1} = W_t - \eta \hat{L}_t^{-1/p} G_t \hat{R}_t^{-1/p}$
- 11: **end for**

**Algorithm 2** Stale Update for Shampoo (left factor)

**Require:** Current factor  $L_t$ ; previous cached state  $\mathcal{S}_{t-1}^L$ ; previous damping  $\epsilon_{t-1}^L$ ; current iteration  $t$

**Hyperparameter:** Stale frequency  $\mathbf{f}$ ; base damping  $\epsilon_0$

- 1: **if**  $(t - 1) \bmod \mathbf{f} = 0$  **then**
- 2:   */\* Refresh once every  $\mathbf{f}$  iterations \*/*
- 3:   Compute  $Q_L D_L Q_L^\top = \text{EVD}(L_t)$
- 4:    $\hat{L}_t^{-1/p} \leftarrow Q_L (D_L + \epsilon_0 \mathbf{I}_m)^{-1/p} Q_L^\top$
- 5:    $\mathcal{S}_t^L \leftarrow (Q_L, D_L)$
- 6: **else**
- 7:   */\* No refresh: reuse prev. inverse-root factor \*/*
- 8:    $\hat{L}_t^{-1/p} \leftarrow \hat{L}_{t-1}^{-1/p}$
- 9:    $\mathcal{S}_t^L \leftarrow \mathcal{S}_{t-1}^L$
- 10: **end if**
- 11:  $\epsilon_t^L \leftarrow \epsilon_0$
- 12: **return**  $(\hat{L}_t^{-1/p}, \mathcal{S}_t^L, \epsilon_t^L)$

For a symmetric matrix  $A$ , we denote its eigenvalues in descending order by  $\{\lambda_i(A)\}_{i=1}^n$ . The Kronecker product of  $A$  and  $B$  is denoted by  $A \otimes B$ . We use the Loewner order  $A \succeq B$  to indicate that  $A - B$  is positive semidefinite. For a symmetric positive definite matrix  $A$  and  $\alpha \in \mathbb{R}$ , the matrix power is defined via eigendecomposition ( $\text{EVD}(A)$ ) as  $A^\alpha := P D^\alpha P^\top$ , where  $A = P D P^\top$ .

**Shampoo.** Shampoo optimizer<sup>2</sup>, presented as in [Algorithm 1](#), provides a preconditioning matrix-valued parameter  $W_t \in \mathbb{R}^{m \times n}$ . It maintains the Kronecker-factored second-moment statistics ( $L_t$  and  $R_t$ ) via exponential moving-average updates of the gradients with parameter  $\beta$ . This

<sup>2</sup>Depending on the choice of the root power  $p$ , this framework covers both the original Shampoo for  $p = 4$  ([Gupta et al., 2018](#)) and Shampoo<sup>2</sup> for  $p = 2$  ([Morwani et al., 2025](#)).

code is written to be a flexible execution loop, with the specific preconditioning strategy encapsulated in an interchangeable  $\mathcal{U}(\cdot)$  procedure highlighted in the blue box. The output of  $\mathcal{U}(\cdot)$  consists of inverse-root factors applied to the gradient  $G_t$  for the parameter update. Via the Kronecker identity, the update becomes  $w_{t+1} = w_t - \eta(\hat{L}_t^{-1/p} \otimes \hat{R}_t^{-1/p})g_t$ , where  $w_t := \text{vec}(W_t)$  and  $g_t := \text{vec}(G_t)$ .

**Stale update.** As a baseline instance of the generalized framework described in [Algorithm 1](#), the Stale update follows a fixed frequency schedule without any auxiliary execution conditions (please see [Algorithm 2](#)). Under this configuration, the update routine is invoked unconditionally every  $\mathbf{f}$  steps to perform a full eigendecomposition (EVD) for both  $L_t$  and  $R_t$ . Following the EVD, a fixed base damping  $\epsilon_0$  is applied to construct the inverse root factors  $\hat{L}_t^{-1/p}$  and  $\hat{R}_t^{-1/p}$ . These factors are then returned to the main skeleton for periodic reuse. For brevity, we state the stale update only for the left Kronecker factor  $L_t$ . The right factor  $R_t$  is updated by the same rule with the substitutions  $L \mapsto R, m \mapsto n, Q_L \mapsto Q_R, D_L \mapsto D_R$ , and  $\epsilon_t^L \mapsto \epsilon_t^R$ .

## 4. Theoretical Objectives: Stability and Regret

We introduce the central quantities we analyze and control: the staleness-induced operator gaps and discounted regret.

### 4.1. Objective 1: Staleness-induced Preconditioner Gap

This section formalizes the concept of the preconditioner gap, defined as the discrepancy  $P_t - \hat{P}_t$  between the ideal fresh preconditioner and its stale counterpart. From this error, we derive the operator gaps,  $\Delta_L$  and  $\Delta_R$ , which represent the operator gap within each Kronecker inverse-root factor.

**Preconditioner gap.** Let  $C_{\mathbf{f}} = \{1 + k\mathbf{f} : k \geq 0\}$  denote the check times, and let  $\mathcal{C} \subseteq \{1, \dots, T\}$  denote the set of iterations at which a fresh EVD is performed. Define the most recent EVD time by  $\tau(t) := \max\{s \leq t : s \in \mathcal{C}\}$ . For stale Shampoo,  $\mathcal{C} = C_{\mathbf{f}}$ . We define the fresh and stale preconditioners as:

$$P_t := (L_t + \epsilon_t \mathbf{I}_m)^{-1/p} \otimes (R_t + \epsilon_t \mathbf{I}_n)^{-1/p},$$

$$\hat{P}_t := \hat{L}_t^{-1/p} \otimes \hat{R}_t^{-1/p},$$

where  $\hat{L}_t := L_{\tau(t)}$  and  $(\hat{L}_t)^{-1/p} := (L_{\tau(t)} + \epsilon_t \mathbf{I}_m)^{-1/p}$  are the stale Kronecker factors. We call  $L_t - \hat{L}_t$  drift. The same applies to  $\hat{R}_t, (\hat{R}_t)^{-1/p}$ .

**Preconditioner gap decomposition.** To analyze how staleness affects the update  $w_{t+1} = w_t - \eta \hat{P}_t g_t$ , we decompose the preconditioner gap into its constituent components using Kronecker product identities:

$$P_t - \hat{P}_t = \hat{L}_t^{-1/p} \otimes \Delta_R(t) + \Delta_L(t) \otimes \hat{R}_t^{-1/p} + \Delta_L(t) \otimes \Delta_R(t).$$

A detailed derivation of this decomposition is provided in [Appendix D](#). In this expression, we define the individual operator errors for each factor as:

$$\begin{aligned}\Delta_L(t) &:= (L_t + \epsilon_t \mathbf{I}_m)^{-1/p} - \hat{L}_t^{-1/p}, \\ \Delta_R(t) &:= (R_t + \epsilon_t \mathbf{I}_n)^{-1/p} - \hat{R}_t^{-1/p}.\end{aligned}$$

## 4.2. Objective 2: Discounted Regret

As the second core metric, we analyze the discounted regret to evaluate convergence performance under temporal staleness. The definition of the discounted regret is:

$$\sum_{t=1}^T \beta^{T-t} f_t(W_t) - \sum_{t=1}^T \beta^{T-t} f_t(W^*),$$

where  $\beta \in (0, 1)$  denotes the discount factor and  $W^*$  is the comparator. Unlike standard regret, this formulation prioritizes more recent performance, which is essential for capturing the dynamic nature of an optimizer that reuses stale statistics.

## 5. Staleness Error and Its Inhibitor $\epsilon$

In this section, we derive mathematical upper bounds for our dual objectives and analyze their physical implications. We show that increasing the refresh period  $\mathbf{f}$  amplifies the staleness error, which we categorize into two distinct forms:

- **Operator-gap staleness error:** Quantifies the operator gap  $\Delta$  that undermines preconditioning stability.
- **Regret staleness error:** Measures the cumulative impact of stale statistics on convergence performance.

While the damping parameter  $\epsilon$  serves as a primary inhibitor of these errors, it systematically suppresses the instability induced by delayed updates ([Section 5.1](#) and [5.2](#)). However, there is a critical trade-off: while a larger  $\epsilon$  suppresses the staleness-related operator error, it also weakens the preconditioning effect. When  $\epsilon$  is excessively large, the inverse-root factors become overly damped and less responsive to the curvature information encoded in the Kronecker factors, thereby reducing the benefit of preconditioning ([Section 5.2](#)).

### 5.1. Operator-gap Staleness Error

**Stale statistics.** The first step in our analysis is to quantify the optimizer’s operator gap. This is measured by the discrepancy between the true preconditioner and its stale versions. We define the filtration  $\mathcal{F}_t := \sigma(W_1, G_1, \dots, G_t)$ . Our analysis relies on the following assumption about the stochastic gradient noise.

**Assumption 5.1.** For some constants  $R > 0$  and  $\sigma > 0$ , the conditional mean satisfies  $\|\mathbb{E}[G_t | \mathcal{F}_{t-1}]\|_2 \leq R$ , and the centered noise is conditionally sub-Gaussian in spectral norm: for all  $u \geq 0$  and all  $1 \leq t \leq T$ :

$$\mathbb{P}\left(\|G_t - \mathbb{E}[G_t | \mathcal{F}_{t-1}]\|_2 \geq u \mid \mathcal{F}_{t-1}\right) \leq 2 \exp\left(-\frac{u^2}{2\sigma^2}\right).$$

Under these conditions, we derive a high-probability bound on the operator gap, *i.e.*,  $\|\Delta_L(t)\|$  and  $\|\Delta_R(t)\|$ , making the dependence on the refresh period  $\mathbf{f}$  and damping  $\epsilon$  explicit. By analogous arguments for the right factor, the full analysis for  $\Delta_R(t)$  is deferred to [Appendix E](#). Henceforth, we focus on the left factor for brevity.

**Lemma 5.2 (Operator Gap Bound).** *Under Assumption 5.1, for any  $\epsilon > 0$  and any  $p \geq 1$ , with probability at least  $1 - \delta$ , the following holds simultaneously for all  $1 \leq t \leq T$ :*

$$\begin{aligned}\|\Delta_L(t)\|_2 &\leq \frac{1}{p\epsilon^{(p+1)/p}} \|L_t - \hat{L}_t\|_2 \\ &\leq \underbrace{\frac{2}{p\epsilon^{(p+1)/p}} (1 - \beta^{\mathbf{f}}) R_{\text{SG}}^2}_{\text{Operator-gap staleness error}},\end{aligned}\tag{1}$$

where  $R_{\text{SG}} := R + K\sigma\sqrt{\log(T/\delta)}$  for a universal constant  $K > 0$ .

Refer to [Appendix E](#) for the proof. The bound in (1) highlights two distinct mechanisms that drive operator error.

**Lipschitz instability (inverse-root sensitivity).** The first inequality isolates the sensitivity of the inverse-root map through the factor  $1/(p\epsilon^{(p+1)/p})$  (cf. [Lemma C.1](#)). This term is an upper bound on the Lipschitz constant; as  $\epsilon \rightarrow 0$ , the map becomes increasingly sensitive, so even small drift can induce a large operator gap. Conversely, increasing  $\epsilon$  reduces this sensitivity and attenuates the effect of large drift on the inverse-root factors. Thus,  $\epsilon$  acts not only as a numerical stabilizer but also as a control variable for operator stability under staleness. We also analyze spectral stability under staleness, including the eigenvalue gap and eigenspace rotation in [Appendix F](#). Importantly, both effects inherit the same  $\epsilon$ -dependent sensitivity of the inverse-root map.

**EMA evolution across a stale window.** The second inequality quantifies how stale statistics lag behind fresh ones over a window of length  $\mathbf{f}$ , as captured by  $(1 - \beta^{\mathbf{f}})$ . When  $\beta$  is close to 1,  $(1 - \beta^{\mathbf{f}}) \approx \mathbf{f}(1 - \beta)$ , skipping refreshes yields a nearly linear growth of drift with  $\mathbf{f}$ . This is the predictable component of operator-gap staleness error.

**$\epsilon$  as a staleness error inhibitor.** Together, these effects reveal a fundamental relationship: while increasing  $\mathbf{f}$  reduces computational overhead, it inherently amplifies the



staleness error. Within this framework,  $\epsilon$  serves as a critical inhibitor, mitigating the sensitivity of the inverse-root factors to ensure stable operator gap bounds. This interaction constitutes the local manifestation of the staleness error.

## 5.2. Regret Staleness Error

To determine the effect of operator error on the regret bound, we derive a discounted regret bound for stale Shampoo. This bound makes explicit how staleness enters the regret through a term that depends on the refresh interval  $\mathbf{f}$ . It therefore provides a quantitative characterization of the contribution of stale updates to the regret bound. At the same time, the bound shows that an excessively large  $\epsilon$  weakens the benefit of preconditioning, revealing a trade-off between error suppression and preconditioning effectiveness. For results for stale Shampoo with  $p = 4$  and the full proof, see Appendix G.

**Assumption 5.3.** The loss  $f_t(\cdot)$  is convex for all  $1 \leq t \leq T$ .

**Theorem 5.4** (Discounted Regret Bound). *Assume Assumption 5.1 and Assumption 5.3. Furthermore, assume  $\text{rank}(G_t) \leq r \leq \min(m, n)$  and  $\max_{t \in T} \|W_t - W^*\|_F = D(W^*) =: D$  for all  $W^* \in \mathbb{R}^{m \times n}$ . Let us denote the constant damping in stale Shampoo as  $\epsilon \equiv \epsilon_t$  for all  $t$ . Then, with probability at least  $1 - \delta$ , the discounted regret of stale Shampoo with  $p = 2$  satisfies*

$$\begin{aligned} & \sum_{t=1}^T \beta^{T-t} f_t(W_t) - \sum_{t=1}^T \beta^{T-t} f_t(W^*) \\ & \leq \frac{1}{2\eta} \left\{ \epsilon D^2 + \frac{\sqrt{R_{\text{SG}}^2 + \epsilon}}{\sqrt{\epsilon}} D^2 R_{\text{SG}}^2 + \frac{D^2 (R_{\text{SG}}^2 + \epsilon)}{\beta} \right\} \\ & \quad + \underbrace{\frac{\eta}{1-\beta} \left\{ \frac{r(1-\beta^{\mathbf{f}}) R_{\text{SG}}^4}{\epsilon^2} \right\}}_{\text{Regret staleness error}} \\ & \quad + \frac{\eta r m n}{2(1-\beta)} \left\{ \log \left( \epsilon + \frac{(1-\beta^T) r R_{\text{SG}}^2}{m n} \right) - \log(\epsilon) \right\}, \end{aligned} \quad (2)$$

where  $R_{\text{SG}} = R + K\sigma\sqrt{\log(T/\delta)}$  and  $K > 0$  is a universal constant.

**Validation of the stability logic.** The regret staleness error term in (2) scales linearly with the drift factor  $(1 - \beta^{\mathbf{f}})$  and inversely with  $\epsilon^2$ . Thus, as the refresh period  $\mathbf{f}$  increases to reduce computational overhead,  $\epsilon$  must increase to prevent staleness-driven error accumulation.

**The risk of over-inhibition.** The regret bound also reveals a countervailing constraint not visible at the operator level: the leading term includes  $\epsilon D^2$ , reflecting the base cost of damping. Hence, overly large  $\epsilon$  inflates the regret through the distance component, producing an intrinsic ceiling on admissible damping. Overall, stable and efficient stale Shampoo requires controlling the operator gap while tuning  $\epsilon$  to balance inverse-root stability against preconditioning strength, rather than fixing it a priori.

## 6. Proposed Method: FOAM Update

To address the staleness error —as established in Section 5—we propose the Frequency and Operator-error based Adaptive Damping Method (FOAM). The key idea is to treat the spectral refresh frequency and the damping level  $\epsilon$  as *coupled* control variables. In the following, we detail the procedure of FOAM.

**Algorithm 3** FOAM factor-wise update rule at iteration  $t$  (left factor)

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**Require:**  $L_t; \mathcal{S}_{t-1}^L = (Q_L, D_L); \hat{L}_{t-1}^{-1/p}; \epsilon_{t-1}^L; t$   
**Hyperparameter:** Check period  $\mathbf{f}$ ; max damping  $\epsilon_{\max}$ ; base damping  $\epsilon_0 \in (0, \epsilon_{\max})$ ; threshold  $\tau \in (0, 1)$

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1: if  $t = 1$  then
2:   Compute  $Q_L D_L Q_L^\top = \text{EVD}(L_t)$ 
3:    $\epsilon_t^L \leftarrow \epsilon_0$ 
4:    $\hat{L}_t^{-1/p} \leftarrow Q_L (D_L + \epsilon_t^L \mathbf{I}_m)^{-1/p} Q_L^\top$ 
5:    $\mathcal{S}_t^L \leftarrow (Q_L, D_L)$ 
6: else if  $(t-1) \bmod \mathbf{f} = 0$  then
7:   /* 1. Compute error proxy */
8:   Compute  $h_t^L = \text{RC}_t^L(\epsilon_{t-1}^L) \cdot \alpha_t^L(\epsilon_{t-1}^L)/p$ 
9:   /* 2. Update damping */
10:   $\tilde{\epsilon}_t^L \leftarrow \max(\epsilon_0, \epsilon_{t-1}^L \frac{h_t^L}{\tau})$ 
11:  if  $\tilde{\epsilon}_t^L \leq \epsilon_{\max}$  then
12:    /* No EVD: Only apply updated damping */
13:     $\epsilon_t^L \leftarrow \tilde{\epsilon}_t^L$ 
14:     $\hat{L}_t^{-1/p} \leftarrow Q_L (D_L + \epsilon_t^L \mathbf{I}_m)^{-1/p} Q_L^\top$ 
15:     $\mathcal{S}_t^L \leftarrow \mathcal{S}_{t-1}^L$ 
16:  else
17:    /* Refresh by EVD and reset damping */
18:    Compute  $Q_L D_L Q_L^\top = \text{EVD}(L_t)$ 
19:     $\epsilon_t^L \leftarrow \epsilon_0$ 
20:     $\hat{L}_t^{-1/p} \leftarrow Q_L (D_L + \epsilon_t^L \mathbf{I}_m)^{-1/p} Q_L^\top$ 
21:     $\mathcal{S}_t^L \leftarrow (Q_L, D_L)$ 
22:  end if
23: else
24:   $\hat{L}_t^{-1/p} \leftarrow \hat{L}_{t-1}^{-1/p}$ 
25:   $\epsilon_t^L \leftarrow \epsilon_{t-1}^L$ 
26:   $\mathcal{S}_t^L \leftarrow \mathcal{S}_{t-1}^L$ 
27: end if
28: return  $(\hat{L}_t^{-1/p}, \mathcal{S}_t^L, \epsilon_t^L)$ 

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### 6.1. Update Inverse-root Factors for Shampoo via FOAM

Sections 4–5 analyze stale Shampoo under a shared damping and fixed refresh schedule and motivate controlling inverse-root sensitivity via damping. FOAM extends this principle to a practical factor-wise controller with adaptive damping and factor-wise refresh decisions.

The update mechanism consists of three phases, summarized in Algorithm 3. For brevity, we state the update only for the left Kronecker factor  $L_t$ . The right factor  $R_t$  is updated

by the same rule with the substitutions  $L \mapsto R$ ,  $m \mapsto n$ ,  $Q_L \mapsto Q_R$ ,  $D_L \mapsto D_R$ , and  $\epsilon_t^L \mapsto \epsilon_t^R$ . For the remainder of Section 6, we use one-factor notation  $h_t, \epsilon_t, \hat{L}_t$  for brevity; the right factor follows by the corresponding substitutions.

**Phase 1: Sensing (operator gap estimation).** This phase estimates whether the current damping level  $\epsilon_{t-1}$  is sufficient to suppress the operator gap induced by the reuse. Rather than computing a fresh eigendecomposition to evaluate the true inverse-root operator gap, FOAM provides a proxy:

$$h_t = \frac{\alpha(\epsilon_{t-1})}{p} \text{RC}(\epsilon_{t-1}),$$

where  $\text{RC}(\cdot)/p$  measures the mismatch between the current inverse preconditioners and the stale ones, and  $\alpha(\epsilon)$  rescales this mismatch into a tighter proxy. Detailed definitions and the theoretical derivation of these terms are provided in Section 6.2

**Phase 2: Adapting (dynamic damping).** Given the proxy  $h_t$ , FOAM enforces the stability condition derived in Section 5.1 via a multiplicative feedback rule:

$$\epsilon_t \leftarrow \max\left(\epsilon_0, \epsilon_{t-1} \cdot \frac{h_t}{\tau}\right).$$

If the estimated error  $h_t$  exceeds a threshold  $\tau$ , we increase  $\epsilon_t$  to stabilize the update. If  $h_t \leq \tau$ , we decrease  $\epsilon_t$  towards its baseline  $\epsilon_0$ . Effectively,  $\epsilon_t$  serves as an inhibitor that suppresses staleness errors. This allows FOAM to reuse old statistics efficiently while keeping the total error within a manageable range.

**Phase 3: Frequency control (triggered refresh).** While increasing  $\epsilon$  helps stability, too much of it can reduce the benefits of the optimizer, as analyzed in Section 5.2. To prevent this, FOAM sets a maximum limit,  $\epsilon_{\max}$ . We continue to reuse old statistics as long as  $\epsilon_t$  stays below this limit. However, if the error becomes so large that  $\epsilon_t$  exceeds  $\epsilon_{\max}$ , FOAM triggers a full refresh of the statistics and resets  $\epsilon_t$  to its baseline. This design ensures we perform EVD only when damping is no longer sufficient to maintain stability.

## 6.2. Construction of the Error Proxy $h_t$

We now introduce the error proxy  $h_t$ , a tractable surrogate designed to estimate the operator gap  $\Delta$  using only the available statistics. It serves as a sensor that monitors operator error caused by staleness to determine when  $\epsilon$  must be adjusted to maintain the stability of the algorithm. The design of  $h_t$  is governed by three primary considerations: grounding in perturbation theory, actionable control logic for  $\epsilon$ , and computational tractability.

**Relative perturbation as a stability metric.** The decision to refresh a preconditioner should be based not on the absolute reuse error  $\|\Delta_L(t)\|_F$ , but on the relative changes it

induces in the preconditioner- *i.e.*,  $\|\Delta_L(t)\|_F$  normalized by the norm of the stale preconditioner currently in use. Drawing from matrix perturbation theory (Horn & Johnson, 2012; Bhatia, 2013), we define our sensing objective as controlling the relative inverse-root operator gap:  $\frac{\|\Delta_L(t)\|_F}{\|\hat{L}_t^{-1/p}\|_F}$ .

Unlike standard diagonalization residuals (DR-Shampoo, (Eschenhagen et al., 2026)), which mainly track how well an eigenspace diagonalizes the current statistic, the relative gap measures the fidelity of the inverse-root mapping itself. This mapping is anisotropic: it is far more sensitive to perturbations that act on low-spectrum directions, where the inverse-root can strongly amplify small drifts. By normalizing the magnitude of the damped operator in use, the relative operator gap yields a health signal that is more directly tied to the stability of the Shampoo update (refer to Appendix I for a detailed comparison). We also present a synthetic experiment in Appendix K that shows our proxy is a better approximation of the true relative error than the diagonalization error used in Eschenhagen et al. (2026).

**Multiplicative damping via scaling logic.** We use the proxy  $h_t$  as a feedback signal to update damping at check steps. The key point is that  $h_t$  depends on  $\epsilon_t$  through the whitening factors  $\hat{L}_t^{-1/2} = (\hat{L}_t + \epsilon_t \mathbf{I}_m)^{-1/2}$ : increasing  $\epsilon_t$  down-weights low-eigenvalue directions, precisely where inverse-root error amplification is strongest, and therefore *reduces* the relative operator-error signal captured by  $\text{RC}(\epsilon)$ . This motivates a feedback update (with a floor at  $\epsilon_0$ ): if  $h_t > \tau$  we increase  $\epsilon_t$  to suppress sensitivity and restore stability, whereas if  $h_t \leq \tau$  we relax  $\epsilon_t$  toward  $\epsilon_0$  to preserve preconditioning strength.

**Derivation of the tractable relative operator error.** While the exact relative operator gap is the ideal metric, computing it precisely would require an EVD, which defeats the purpose of our method. We therefore seek a proxy that (i) is computable in the stale eigenspaces and (ii) upper-bounds the relative change induced by staleness. To this end, we approximate the relative operator error and analyze the resulting first-order term. We obtain a bound on the *relative* magnitude of the first-order approximation  $\Delta_L^{(1)}$ :

$$\frac{\|\Delta_L^{(1)}\|_F}{\|\hat{L}_t^{-1/p}\|_F} \leq \frac{\alpha}{p} \left\| (\hat{L}_t^{-1/2})(L_t - \hat{L}_t)(\hat{L}_t^{-1/2}) \right\|_F = h_t \quad (3)$$

where  $\alpha(\epsilon) := \|\hat{L}_t^{-1/p}\|_2 / \|\hat{L}_t^{-1/p}\|_F$  and  $\text{RC}(\epsilon_{t-1}) := \|(\hat{L}_t^{-1/2})(L_t - \hat{L}_t)(\hat{L}_t^{-1/2})\|_F$ . A complete proof of (3) is provided in Appendix H.

## 7. Experiments

In this section, we evaluate the performance of the proposed algorithm across several deep learning tasks.

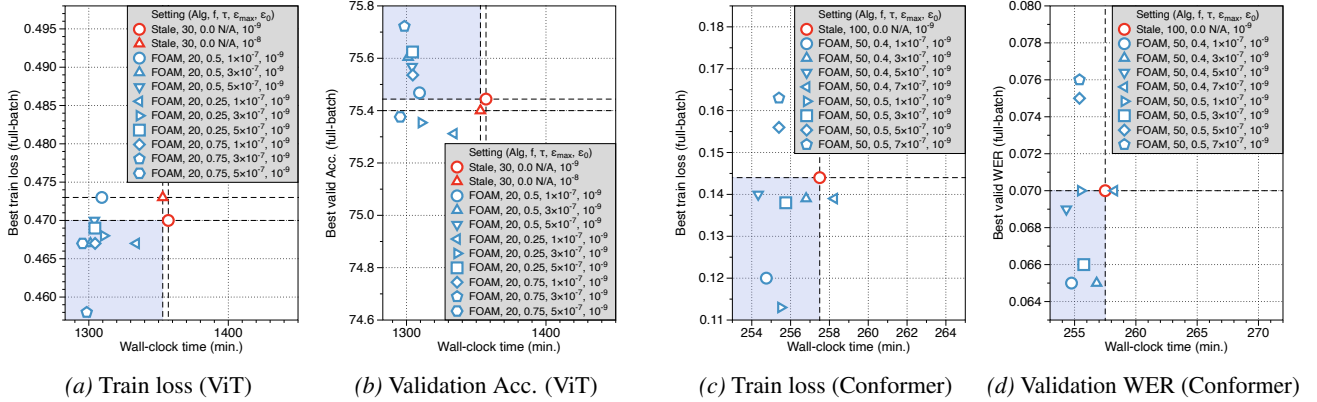


Figure 1. Wall-clock efficiency comparison between Shampoo with stale update and FOAM update. Figure 1a-1b presents the best training loss and validation accuracy for the ViT (ImageNet-1K) task, while Figure 1c-1d shows the training loss and WER for the Conformer (LibriSpeech) task. Across various combinations of the two introduced hyperparameters ( $\tau, \epsilon_{\max}$ ), FOAM improves the wall-clock/performance trade-off over stale Shampoo.

## 7.1. Experimental Setup

This section details the experimental configurations and benchmarking protocols used to evaluate the wall-clock efficiency and robustness of FOAM in diverse tasks.

**Tasks and benchmarks.** We evaluate the performance of FOAM on three benchmarks: (1) image classification on ImageNet-1K (Deng et al., 2009) using a Vision Transformer (ViT-small) (?), (2) automatic speech recognition (ASR) on LibriSpeech (Panayotov et al., 2015) using a Conformer (Gulati et al., 2020), and (3) language modeling on Wikitext-103-v1 (Merity et al., 2016) using GPT-2 (Radford et al., 2019).

**Implementation details.** We have provided detailed information in the Appendix J regarding the hardware and hyperparameter settings for each experiment, as well as other details necessary for reproducibility.

## 7.2. Experimental Results

First, in ViT, FOAM exhibits a significant wall-clock advantage while matching or exceeding the performance of other baselines, as illustrated in Table 1. A similar trend is observed in the Conformer experiments Table 2, where FOAM achieves a lower WER in less time. The blue rectangular regions in each plot highlight the area of superior performance in which FOAM resides.

Furthermore, these gains are consistent across a broad range of hyperparameters as illustrated in Figure 1. A comprehensive sweep over  $(\tau, \epsilon_{\max})$  shows that the majority of tested configurations fall within the superior blue regions, demonstrating that the wall-clock advantage of FOAM is robust and independent of a single tuned setting. This stability confirms the effectiveness of the proxy-driven controller in maintaining performance across various training scenarios.

## 7.3. Analysis

We provide a detailed examination of FOAM’s internal dynamics, investigating the evolution of the adaptive damping parameter  $\epsilon$ , the relative reduction in EVD frequency for Kronecker factors, and the performance impact of its core components through systematic ablation studies on ViT. For these analysis, the hyperparameter configurations are set to  $(f, \tau, \epsilon_{\max}, \epsilon_0) = (20, 0.75, 3 \times 10^{-7}, 10^{-9})$  for ViT and  $(50, 0.4, 1 \times 10^{-7}, 10^{-9})$  for Conformer. Other hyperparameters are referred to in Appendix J.

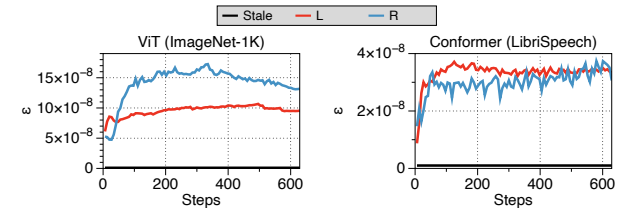


Figure 2.  $\epsilon$  dynamics over iterations.

**The evolution of adaptive damping  $\epsilon$ .** Figure 2 illustrates the adaptive  $\epsilon$  dynamics of FOAM, contrasting it with the static damping maintained by the stale update baseline. While the stale baseline relies on a fixed, minimal value, FOAM adaptively raises  $\epsilon_t$  as a compensatory mechanism to counteract spectral drift and improves numerical robustness during the sensing phase. This helps maintain stable optimization behavior even as the age of the Kronecker factors increases. Also, to prevent the side effect of over-damping — which can degrade preconditioning quality — the controller incorporates a modulation strategy that regulates the growth and relaxation of  $\epsilon_t$ .

**EVD frequency.** Figure 3 quantifies the reduction in computational overhead by comparing the EVD frequency of FOAM against the fixed-cadence stale update method, normalized to 100%. While the stale baseline performs EVD

| Optimizer   | $(f, \tau, \epsilon_0, \epsilon_{\max})$    | Train Loss | Validation accuracy | Wall-clock Time |
|-------------|---------------------------------------------|------------|---------------------|-----------------|
| AdamW       | (N/A, N/A, $10^{-9}$ , N/A)                 | 0.635      | 72.39 %             | 1,258 (minute)  |
| Shampoo     | (20, N/A, $10^{-9}$ , N/A)                  | 0.498      | 75.13%              | 1,415 (minute)  |
| DR-Shampoo  | (20, 0.75, $10^{-9}$ , N/A)                 | 0.531      | 73.56 %             | 1,310 (minute)  |
| FOAM (Ours) | (20, 0.75, $10^{-9}$ , $3 \times 10^{-7}$ ) | 0.458      | 75.72 %             | 1,300 (minute)  |
| SOAP        | (20, N/A, $10^{-9}$ , N/A)                  | 0.485      | 75.1%               | 1,360 (minute)  |

Table 1. Comparison of wall-clock time and training loss on the ViT task.

| Optimizer   | $(f, \tau, \epsilon_0, \epsilon_{\max})$   | Train Loss | Validation WER | Wall-clock Time |
|-------------|--------------------------------------------|------------|----------------|-----------------|
| AdamW       | (N/A, N/A, $10^{-9}$ , N/A)                | 0.42       | 0.099          | 205 (minute)    |
| Shampoo     | (50, N/A, $10^{-9}$ , N/A)                 | 0.17       | 0.069          | 264 (minute)    |
| DR-Shampoo  | (50, 0.4, $10^{-9}$ , N/A)                 | 0.18       | 0.065          | 294 (minute)    |
| FOAM (Ours) | (50, 0.4, $10^{-9}$ , $3 \times 10^{-7}$ ) | 0.14       | 0.065          | 257 (minute)    |
| SOAP        | (50, N/A, $10^{-9}$ , N/A)                 | 0.17       | 0.067          | 265 (minute)    |

Table 2. Comparison of wall-clock time and training loss on the Conformer task.

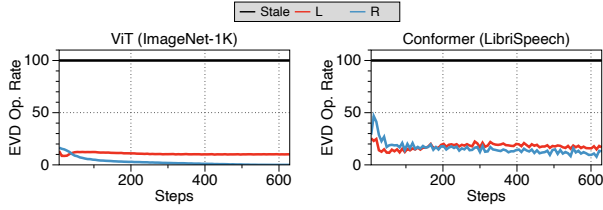


Figure 3. Relative rate of EVD function calls.

updates at every scheduled interval regardless of necessity, FOAM triggers refreshes only when the error proxy  $h_t$  indicates significant spectral drift. In the ViT case, the number of EVD for  $L$  and  $R$  drops to approximately 10% and 5%, respectively. A similar trend is observed in the Conformer benchmark. In addition, we profiled the optimizer wall-clock time and compared the time taken to compute the proxy  $h_t$  in FOAM with the time taken to EVD, reporting the results in the Table 5. In conclusion, the proxy computation was negligible compared to EVD, and given that FOAM performs significantly fewer EVD than stale Shampoo, our wall-clock time gain is attributable to the reduced number of EVD.

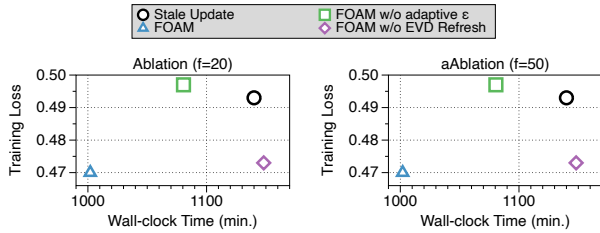


Figure 4. Ablation study of FOAM.

**Ablation study.** Figure 4 presents an ablation study on the ViT architecture, demonstrating the critical synergy between adaptive damping and selective EVD refreshes. When

increasing the interval  $f$  to gain computational efficiency without utilizing adaptive  $\epsilon$  compensation (green square), the training loss increases significantly as the stale preconditioner fails to account for spectral drift. Conversely, relying solely on  $\epsilon$  control without triggering EVD refreshes (purple diamond) can reduce the training loss, but at the cost of significantly higher wall-clock time. Only the full configuration of FOAM (blue triangle) achieves the balance, simultaneously minimizing both training loss and computational overhead. These results confirm that integrating proxy-driven damping adaptation and well-timed refreshes is essential for maintaining high performance within a reduced wall-clock budget.

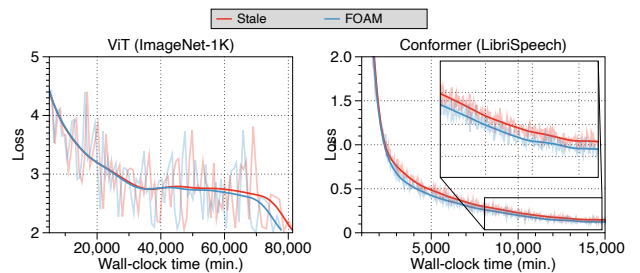


Figure 5. Learning curve. (Wall-clock time vs Loss)

**Learning curve.** As shown in Figure 5, FOAM consistently achieves lower training loss than Stale across both ViT and Conformer. The magnified insets further reveal that FOAM maintains superior convergence, confirming that its mechanism preserves and stabilizes the optimization while operating efficiently.



## 8. Conclusion

We presented FOAM, a mathematically grounded framework that resolves the heuristic choice of stale Shampoo. By introducing an error proxy and an adaptive damping feedback controller, our method enables stable optimization behavior while reducing the frequency of EVD relative to stale Shampoo. Our extensive evaluation provides evidence that FOAM consistently achieves significant wall-clock speedups. Ultimately, FOAM bridges the gap between the superior optimization trajectories of Shampoo-style methods and the efficiency required for large-scale deep learning.

## Acknowledgements

This work was supported in part by the Institute of Information & Communications Technology Planning & Evaluation (IITP) grant funded by the Korean government (RS-2025-25464461), and in part by the Korea Evaluation Institute of Industrial Technology (KEIT) grant (RS-2025-02653102).

## Impact Statement

This paper presents FOAM, which aims to improve the computational efficiency and numerical robustness of preconditioned optimization in large-scale machine learning. By providing a theoretical foundation for managing curvature staleness through adaptive damping, this work aims to bridge the gap between high-fidelity optimization and practical training speed-ups. The broader impact of this research is twofold:

- **Resource Efficiency:** By accelerating convergence and reducing hyperparameter sensitivity, our approach contributes to the sustainable development of AI, potentially lowering the energy consumption and carbon footprint associated with training large models.
- **Training Reliability:** Enhancing the stability of the preconditioned method is crucial for developing more reliable and trustworthy AI systems, as it mitigates the risk of training instability during resource-intensive training runs.

We believe that making sophisticated optimization techniques more accessible and stable is essential for the continued progress of the machine learning community.

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**–Supplementary–**  
**FOAM: Frequency and Operator Error-Based Adaptive Damping Method**  
**for Reducing Staleness-Oriented Error for Shampoo**

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## A. Limitations and Future Works

While FOAM demonstrates robustness and efficiency, several avenues for improvement and extension remain:

- **Theoretical extensions to realistic settings:** Our current regret analysis focuses on convex settings. Extending this to non-convex (and possibly non-smooth) online learning regimes would provide guarantees that better reflect the dynamics of deep learning (Ahn et al., 2025; 2024; Cutkosky et al., 2023). Furthermore, our operator error bounds rely on sub-Gaussian assumptions; deriving analogous stability results under weaker conditions, such as bounded variance, represents a crucial step toward bridging the gap between theory and practice.
- **Reducing Hyperparameter complexity:** Experimentally, FOAM has consistently demonstrated superior efficiency and robustness compared to Stale Shampoo across a wide range of  $\epsilon_{\max}$  and  $\tau$ . However, introducing two new hyperparameters expands the search space relative to the original Shampoo. Therefore, research aimed at minimizing the tuning cost for these parameters remains a critical direction for future work.

**Toward FOAM with SOAP-style (eigengap-aware motivation).** SOAP updates use only the eigenbases  $Q_L, Q_R$  of Shampoo’s Kronecker factors to rotate the gradients into an approximate eigenspace, apply an Adam step, and rotate back; the eigenvalues themselves do not explicitly enter the update. Consequently, when  $Q_L, Q_R$  are refreshed only periodically (e.g., via QR-iteration), the dominant staleness effect is not merely a small “operator error” in the factors, but eigenspace drift: stale bases rotate gradients into wrong coordinates, so even a perfectly well-behaved diagonal update is applied along incorrect directions. Standard subspace-perturbation theory shows that the size of this drift is controlled by a ratio of the form “statistics drift/eigengap” (refer to Theorem F.2), so the same amount of change in the Kronecker factors can induce dramatically different basis errors depending on how well-separated the relevant eigenvalues are. This suggests that any Soap with FOAM style analysis should explicitly account for **eigengap-dependent** sensitivity when reasoning about (and bounding) the error induced by stale eigenbases updates.

## B. Notation and Preliminary

### B.1. Notation

The vector space of  $m \times n$  matrices is denoted by  $\mathbb{R}^{m \times n}$ , and the  $m \times m$  zero matrix and identity matrices are denoted by  $\mathbf{0}_m$  and  $\mathbf{I}_m$ , respectively. We denote by  $\mathbb{S}_n$  the set of symmetric  $n \times n$  matrices and by  $\mathbb{S}_{n,+}$  the cone of symmetric positive definite matrices. For  $A \in \mathbb{R}^{m \times n}$ , the trace is  $\text{Tr}(A)$ . The inner product is  $\langle A, B \rangle := \text{Tr}(A^\top B)$ , the Frobenius norm is  $\|A\|_F := \sqrt{\langle A, A \rangle}$ , and the spectral norm is  $\|A\|_2 := \max_{\|x\|_2=1} \|Ax\|_2$ . Given  $H \in \mathbb{S}_{n,+}$ , the energy norm and its dual are

$$\|x\|_H := \sqrt{x^\top H x}, \quad \|x\|_H^* := \sqrt{x^\top H^{-1} x}.$$

For  $A \in \mathbb{S}_n$ , we denote its eigenvalues in descending order by  $\{\lambda_i(A)\}_{i=1}^n$  (or simply  $\{\lambda_i\}_{i=1}^n$  when clear from context). The Kronecker product of  $A \in \mathbb{R}^{m \times n}$  and  $B \in \mathbb{R}^{p \times q}$  is denoted by  $A \otimes B \in \mathbb{R}^{mp \times nq}$ . For  $A \in \mathbb{R}^{m \times n}$  with rows  $a_1, \dots, a_m$ , we define  $\text{vec}(A) := (a_1 \ a_2 \ \dots \ a_m)^\top \in \mathbb{R}^{mn}$ . We write  $A \succeq 0$  (resp.  $A \succ 0$ ) to indicate that  $A$  is positive semidefinite (resp. positive definite). The Loewner order  $A \succeq B$  means  $A - B \succeq 0$ . Finally, for  $A \in \mathbb{S}_{n,+}$  and  $\alpha \in \mathbb{R}$ , the matrix power is defined by  $A^\alpha := PD^\alpha P^\top$ , where  $A = PDP^\top$  is an eigen-decomposition. If  $A$  is merely PSD (not PD),  $A^\alpha$  is well-defined for  $\alpha \geq 0$ . In the Appendix, we denote the base damping factor in Shampoo as  $\epsilon > 0$ .

### B.2. Preliminary

This subsection collects the background tools used in our analysis. We briefly review (i) the interpretation of preconditioning as reparameterization, (ii) discounted regret in Online Convex Optimization (OCO), and (iii) key results from matrix perturbation theory.

**Preconditioned gradient method** Preconditioned gradient methods can be viewed through reparameterization. Let us consider

$$\begin{aligned} \min_{w \in \mathbb{R}^m} \ell(w) \\ w = R\hat{w} \quad \Leftrightarrow \quad \hat{w} = R^{-1}w, \end{aligned}$$

where  $R \in \mathbb{R}^{n \times n}$  is invertible and  $\hat{\ell}(\hat{w}) := \ell(R\hat{w})$  is defined. By the chain rule,

$$\nabla_{\hat{w}} \hat{\ell}(\hat{w}) = R^\top \nabla_w \ell(w).$$

A gradient step in the transformed space,

$$\hat{w}_{t+1} = \hat{w}_t - \eta \nabla_{\hat{w}} \hat{\ell}(\hat{w}_t),$$

corresponds in the original space to

$$w_{t+1} = R\hat{w}_{t+1} = w_t - \eta \underbrace{(RR^\top)}_{=:A} \nabla_w \ell(w_t).$$

Thus, preconditioning by a matrix  $A \succeq 0$  is equivalent to gradient descent after reparameterization via  $R = A^{1/2}$ .

**Online Convex Optimization** We provide a concise overview of online convex optimization (OCO) and discounted regret; see [Orabona \(2019\)](#) for a comprehensive introduction. In OCO, for  $T$  rounds, the learner selects  $w_t \in \mathcal{W} \subseteq \mathbb{R}^n$ , then a convex loss  $f_t : \mathcal{W} \rightarrow \mathbb{R}$  is revealed, and the learner incurs a loss  $f_t(w_t)$ . The (static) regret against any comparator  $w \in \mathcal{W}$  is

$$\text{Reg}_T(w) := \sum_{t=1}^T f_t(w_t) - \sum_{t=1}^T f_t(w).$$

**Discounted regret.** To model nonstationarity and emphasize recent losses, several works consider *discounted regret* ([Ahn et al., 2024](#); [Jacobsen & Cutkosky, 2024](#); [Zhang et al., 2024](#)):

$$\text{DReg}_T(w) := \sum_{t=1}^T \beta^{T-t} f_t(w_t) - \sum_{t=1}^T \beta^{T-t} f_t(w),$$

where  $\beta \in (0, 1]$  is a discount factor. Discounted regret is (i) a valuable surrogate for dynamic regret via discounted-to-dynamic conversion ([Ahn et al., 2024](#)) and (ii) a convenient metric for designing adaptive online algorithms in evolving environments ([Zhang & Cutkosky, 2024](#)).

**Why discounted regret here.** Our stale strategy reuses preconditioner factors over a window, making the effective update rule time-varying and history-dependent. Following the discounted regret framework developed for Adam-type updates ([Ahn et al., 2024](#)), we derive discounted regret bounds for stale Shampoo with  $p = 2$  and stale Shampoo with  $p = 4$ , explicitly capturing the dependence on the staleness frequency  $\mathbf{f}$  and damping  $\epsilon$ .

**Matrix perturbation theory** We briefly review perturbation tools for symmetric matrices to quantify how stale statistics deviate from fresh inverse-root factors and how these deviations propagate to the preconditioner. See [Horn & Johnson \(2012\)](#) for a detailed treatment.

Our analysis distinguishes between (i) the stability of eigenvalues and (ii) the sensitivity of invariant subspaces, and then connects these classical facts to inverse-root matrix functions.

- **Eigenvalue stability (Weyl’s inequality; Lemma C.2).** Let  $A \in \mathbb{S}_n$  and  $E \in \mathbb{S}_n$ , and denote by  $\lambda_1(\cdot) \leq \dots \leq \lambda_n(\cdot)$  the ordered eigenvalues. Weyl’s inequality yields

$$|\lambda_i(A + E) - \lambda_i(A)| \leq \|E\|_2 \quad \text{for all } i \in \{1, \dots, n\}.$$

Thus, in the spectral norm, small perturbations imply uniformly small eigenvalue shifts.

- **Eigenspace sensitivity (Davis–Kahan; Lemma C.3).** In contrast, invariant subspaces can be much more sensitive. Let  $V \in \mathbb{R}^{n \times d}$  be an orthonormal basis for an invariant subspace of  $A$  associated with a target eigenvalue cluster/interval, and let  $\hat{V}$  be the corresponding invariant subspace of  $A + E$ . If this cluster is separated from the rest of the spectrum by an eigengap  $\delta > 0$ , then Davis–Kahan gives a  $\sin \Theta$  bound of the form

$$\|\sin \Theta(V, \hat{V})\|_2 \leq \frac{\|E\|_2}{\delta},$$

(see Lemma C.3 for the precise statement). Consequently, when eigenvalues are clustered (i.e.,  $\delta \downarrow 0$ ), even small perturbations can induce large rotations of the preconditioning basis, despite eigenvalues remaining stable.

- **Worst-case Lipschitz behavior of inverse-root maps and  $\epsilon$ -dependence.** Shampoo requires inverse-root matrix functions of the form

$$F_\epsilon(X) = (X + \epsilon \mathbf{I})^{-1/p}.$$

Over a regime where  $X \succeq 0$  and  $X + \epsilon \mathbf{I} \succ 0$ , Lemma C.1 implies that  $F_\epsilon$  is (globally) Lipschitz, with a worst-case constant controlled by the scalar map  $f(t) = (t + \epsilon)^{-1/p}$  and its derivative

$$f'(t) = -\frac{1}{p}(t + \epsilon)^{-(p+1)/p}.$$

In particular, since  $\sup_{t \geq 0} |f'(t)| = \frac{1}{p}\epsilon^{-(p+1)/p}$ , one obtains the conservative worst-case scaling

$$\text{Lip}(F_\epsilon) \lesssim \frac{1}{p\epsilon^{(p+1)/p}} = \mathcal{O}(\epsilon^{-(p+1)/p}).$$

We refer to this blow-up as *Lipschitz instability*: as  $\epsilon \downarrow 0$ , the worst-case upper bound on  $\|F_\epsilon(X + E) - F_\epsilon(X)\|$  can become arbitrarily large, even for small damping factor  $\|E\|$ .

**Anisotropy and “small-eigenvalue directions.”** Crucially, the above worst-case Lipschitz bound holds for *any* perturbation direction, but it becomes nearly tight only when the drift interacts strongly with the most sensitive low-spectrum directions (eigenvalues near  $\epsilon$ ). This is inherently *anisotropic*: the first-order variation is governed by the Fréchet derivative  $DF_\epsilon(X)[E]$ , whose amplification factors depend on the spectrum of  $X$  through  $f'$  (and divided-difference coefficients in the non-commuting case).

To make the notion of “small-eigenvalue directions” explicit, consider the commuting (simultaneously diagonalizable) case: assume  $XE = EX$  with  $X, E \in \mathbb{S}_n$ , so there exists an orthogonal  $U$  such that

$$U^\top XU = \text{diag}(\lambda_1, \dots, \lambda_n), \quad U^\top EU = \text{diag}(\delta_1, \dots, \delta_n).$$

Define the low-spectrum index set at level  $\tau > 0$  by

$$\mathcal{I}_{\text{low}}(\tau) := \{i : \lambda_i \leq \tau\}, \quad P_{\text{low}}(\tau) := \sum_{i \in \mathcal{I}_{\text{low}}(\tau)} u_i u_i^\top,$$

where  $u_i$  are the eigenvectors of  $X$ . Then “drift aligned with small-eigenvalue directions” can be quantified, e.g., by the concentration ratio

$$\rho_{\text{low}}(E; \tau) := \frac{\|P_{\text{low}}(\tau)EP_{\text{low}}(\tau)\|_F}{\|E\|_F} \in [0, 1],$$

which equals  $(\sum_{i \in \mathcal{I}_{\text{low}}(\tau)} \delta_i^2)^{1/2} / (\sum_{i=1}^n \delta_i^2)^{1/2}$  in this commuting setting. When  $\rho_{\text{low}}(E; \tau) \approx 1$  with  $\tau$  on the order of  $\epsilon$ , perturbations concentrate on the most sensitive low-spectrum subspace and the global worst-case Lipschitz bound is (approximately) attained; when  $\rho_{\text{low}}(E; \tau) \ll 1$ , perturbations largely live in large-eigenvalue directions and the change in  $(X + \epsilon \mathbf{I})^{-1/p}$  is typically much smaller than the worst-case prediction. This geometric viewpoint explains why a single global worst-case constant can be overly pessimistic in practice and motivates the direction-aware *a posteriori* error estimator developed in Section 6.2.

## C. Technical Lemmas

**Lemma C.1** (Lipschitz continuity of a matrix function  $F(X) = X^{1/p}$ ). *Consider a compact domain  $\mathcal{D} \subset \mathbb{S}_{n,+}$  defined as:*

$$\mathcal{D} = \{X \in \mathbb{S}_{n,+} \mid a\mathbf{I}_n \preceq X \preceq b\mathbf{I}_n\},$$

where  $0 < a \leq b < \infty$ . For a given non-zero scalar  $p$ , the matrix function  $F : \mathcal{D} \rightarrow \mathbb{S}_n$  defined by  $F(X) = X^{1/p}$  is Lipschitz continuous with respect to the spectral norm  $\|\cdot\|_2$ . That is, for any  $X, Y \in \mathcal{D}$ , the following inequality holds:

$$\|X^{1/p} - Y^{1/p}\|_2 \leq L\|X - Y\|_2,$$

where the Lipschitz constant  $L$  is given by the maximum magnitude of the derivative of the scalar function  $f(t) = t^{1/p}$  on the interval  $[a, b]$ :

$$L = \sup_{t \in [a, b]} \left| \frac{d}{dt} t^{1/p} \right| = \begin{cases} \frac{1}{p} a^{(1-p)/p} & \text{if } p \geq 1, \\ \frac{1}{p} b^{(1-p)/p} & \text{if } 0 < p < 1, \\ \frac{1}{|p|} a^{(1-p)/p} & \text{if } p < 0. \end{cases}$$

*Proof.* This is a standard result in matrix analysis (e.g., (Bhatia, 2013)).  $\square$

**Lemma C.2** (Weyl's inequality and Dual Weyl's inequality). *Let  $A, B \in \mathbb{S}_n$ , and  $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$  denote the ordered eigenvalues. Weyl's inequality and the dual Weyl's inequality state that:*

$$\lambda_{i+j-1}(A+B) \leq \lambda_i(A) + \lambda_j(B), \quad i, j \geq 1, \quad i+j-1 \leq n, \quad \lambda_{i+j-n}(A+B) \geq \lambda_i(A) + \lambda_j(B), \quad i, j \geq 1, \quad i+j-n \geq 1.$$

Hence, we obtain:

$$|\lambda_i(A) - \lambda_i(B)| \leq \|A - B\|_2, \quad 1 \leq i \leq n.$$

*Proof.* Let  $E := A - B$ , so  $A = B + E$ ; applying Weyl's inequality with  $j = 1$  and the dual Weyl inequality with  $j = n$  gives  $\lambda_i(B) + \lambda_i(E) \leq \lambda_i(A) \leq \lambda_i(B) + \lambda_1(E)$ . Hence, for all  $i$ , we get:

$$|\lambda_i(A) - \lambda_i(B)| \leq \max\{|\lambda_1(E)|, |\lambda_n(E)|\} = \|E\|_2 = \|A - B\|_2. \quad \square$$

**Lemma C.3** (Variant of Davis-Kahan Theorem). *Let  $A, B \in \mathbb{S}_m$ , and eigenvalues  $\hat{\lambda}_1 \geq \dots \geq \hat{\lambda}_m$  and  $\lambda_1 \geq \dots \geq \lambda_m$  respectively. Fix  $1 \leq r \leq s \leq m$  and assume that  $\min(\lambda_{r-1} - \lambda_r, \lambda_s - \lambda_{s+1}) > 0$ , where  $\lambda_0 := \infty$ ,  $\lambda_{m+1} := -\infty$ . Let  $d := s - r + 1$  and let  $V = [v_r, \dots, v_s] \in \mathbb{R}^{m \times d}$ ,  $\hat{V} = [\hat{v}_r, \dots, \hat{v}_s] \in \mathbb{R}^{m \times d}$  have orthonormal columns satisfying  $Av_j = \lambda_j v_j$  and  $B\hat{v}_j = \hat{\lambda}_j \hat{v}_j$ , for  $r \leq j \leq s$ . Then:*

$$\|\sin \Theta(V, \hat{V})\|_F \leq \frac{2 \min(d^{1/2} \|B - A\|_2, \|B - A\|_F)}{\min(\lambda_{r-1} - \lambda_r, \lambda_s - \lambda_{s+1})},$$

where  $\sin \Theta$  is the diagonal matrix of the sines of the principal angles.

*Proof.* See Theorem 2 of Yu et al. (2015).  $\square$

**Lemma C.4** (FTL-BTL (Kalai & Vempala, 2005) with discount weights). *Fix  $\beta \in (0, 1)$  and horizon  $T \in \mathbb{N}$ . Let  $\Phi : \mathcal{S}_{d,+} \rightarrow \mathbb{R}$  be any function (typically convex). For  $t \geq 0$ , define*

$$H_t \in \arg \min_{H \succ 0} \left\{ \Phi(H) + \sum_{s=1}^t \langle (1-\beta) \beta^{T-s} g_s g_s^\top, H^{-1} \rangle_F \right\}.$$

(Equivalently, let  $S_t^{(T)} := (1-\beta) \sum_{s=1}^t \beta^{T-s} g_s g_s^\top$  and  $H_t \in \arg \min_{H \succ 0} \{\langle S_t^{(T)}, H^{-1} \rangle_F + \Phi(H)\}$ .) Then

$$(1-\beta) \sum_{t=1}^T \beta^{T-t} \|g_t\|_{H_t}^2 \leq (1-\beta) \sum_{t=1}^T \beta^{T-t} \|g_t\|_{H_T}^2 + \Phi(H_T) - \Phi(H_0).$$



**Lemma C.5.** Let  $G \in \mathbb{R}^{m \times n}$  be a matrix of rank at most  $r$  and denote  $g = \text{vec}(G)$ . Then:

$$\frac{1}{r}gg^T \preceq \mathbf{I}_m \otimes (G^T G) \quad \text{and} \quad \frac{1}{r}gg^T \preceq (GG^T) \otimes \mathbf{I}_n.$$

*Proof.* This is Lemma 9 of [Gupta et al. \(2017\)](#). □

**Lemma C.6** (Monotonicity of the matrix geometric mean). Let  $A, B, C, D \in \mathbb{S}_+^d$  with

$$A \preceq C, \quad B \preceq D.$$

Then

$$A \# B \preceq C \# D,$$

where

$$A \# B := A^{1/2} (A^{-1/2} B A^{-1/2})^{1/2} A^{1/2}.$$

In particular, if  $X \preceq P$  and  $X \preceq Q$ , then

$$X = X \# X \preceq P \# Q.$$

If moreover  $P$  and  $Q$  commute, then

$$P \# Q = P^{1/2} Q^{1/2}.$$

**Lemma C.7.** Assume that  $G_1, \dots, G_T \in \mathbb{R}^{m \times n}$  are matrices of rank at most  $r$ . Let  $g_t = \text{vec}(G_t)$  for all  $1 \leq t \leq T$ . Then, for all  $\epsilon > 0$  and  $0 < \beta < 1$ :

$$\epsilon \mathbf{I}_{mn} + \frac{1-\beta}{r} \sum_{t=1}^T \beta^{T-t} g_t g_t^T \preceq \left( \epsilon \mathbf{I}_m + (1-\beta) \sum_{t=1}^T \beta^{T-t} G_t G_t^T \right)^{1/2} \otimes \left( \epsilon \mathbf{I}_n + (1-\beta) \sum_{t=1}^T \beta^{T-t} G_t^T G_t \right)^{1/2}.$$

*Proof.* Define

$$A_n := \epsilon \mathbf{I}_n + (1-\beta) \sum_{t=1}^T \beta^{T-t} G_t^T G_t, \quad B_m := \epsilon \mathbf{I}_m + (1-\beta) \sum_{t=1}^T \beta^{T-t} G_t G_t^T,$$

and

$$X := \epsilon \mathbf{I}_{mn} + \frac{1-\beta}{r} \sum_{t=1}^T \beta^{T-t} g_t g_t^T.$$

From Lemma C.5, for each  $t$ ,

$$\frac{1}{r} g_t g_t^T \preceq \mathbf{I}_m \otimes (G_t^T G_t), \quad \frac{1}{r} g_t g_t^T \preceq (G_t G_t^T) \otimes \mathbf{I}_n.$$

Summing with weights  $(1-\beta)\beta^{T-t}$  and adding  $\epsilon \mathbf{I}_{mn}$  gives

$$X \preceq \mathbf{I}_m \otimes A_n =: P, \quad X \preceq B_m \otimes \mathbf{I}_n =: Q.$$

Since  $\epsilon > 0$ , we have  $A_n \succ 0$  and  $B_m \succ 0$ , hence  $P, Q \in \mathbb{S}_+^{mn}$ . Therefore, by Lemma C.6,

$$X = X \# X \preceq P \# Q.$$

Moreover,

$$PQ = (\mathbf{I}_m \otimes A_n)(B_m \otimes \mathbf{I}_n) = B_m \otimes A_n = (B_m \otimes \mathbf{I}_n)(\mathbf{I}_m \otimes A_n) = QP,$$

so  $P$  and  $Q$  commute. Hence

$$P \# Q = P^{1/2} Q^{1/2}.$$

Using Lemma C.8,

$$P^{1/2}Q^{1/2} = (\mathbf{I}_m \otimes A_n)^{1/2}(B_m \otimes \mathbf{I}_n)^{1/2} = (\mathbf{I}_m \otimes A_n^{1/2})(B_m^{1/2} \otimes \mathbf{I}_n) = B_m^{1/2} \otimes A_n^{1/2}.$$

Therefore,

$$X \preceq B_m^{1/2} \otimes A_n^{1/2}.$$

□

**Lemma C.8** (Kronecker product properties). *We summarize well-known properties of the Kronecker product. Let  $A, B, C, D$  be appropriately dimensioned matrices.*

1.  $\|A \otimes B\| = \|A\|\|B\|$ , where  $\|\cdot\|$  is the spectral or Frobenius norm;
2.  $(A \otimes B)^k = A^k \otimes B^k$ , for all  $k \in \mathbb{R}$ , where  $A, B \in \mathbb{S}_{n,+}$ ;
3.  $(A + B) \otimes C = (A \otimes C) + (B \otimes C)$ ;
4.  $k(A \otimes B) = A \otimes (kB) = (kA) \otimes B$ , for all  $k \in \mathbb{R}$ ;
5.  $(A \otimes B)(C \otimes D) = (AC \otimes BD)$ .
6.  $\langle A, B \rangle = \text{vec}(A)^\top \text{vec}(B)$ .

**Lemma C.9** (Operator monotone property). *If  $0 \preceq A \preceq B$ , then  $A^{1/k} \preceq B^{1/k}$  for  $k \geq 1$ .*

**Lemma C.10.** *Given  $A \in \mathbb{S}_{m,+}$ :*

$$\det(A) \leq \left( \frac{\text{Tr}(A)}{m} \right)^m.$$

*Proof.*  $\det(A) = \prod_{i=1}^m \lambda_i(A)$  and  $\text{Tr}(A) = \sum_{i=1}^m \lambda_i(A)$ ; apply the AM-GM inequality. □

**Lemma C.11** (Daleckii–Kreĭn Theorem (Noferini, 2016)). *Let  $X \in \mathbb{S}^n$  have the eigendecomposition  $X = U\Lambda U^\top$  with  $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ , and let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be continuously differentiable on an open interval containing the spectrum  $\{\lambda_i\}_{i=1}^n$ . Define the spectral matrix function  $f(X) := Uf(\Lambda)U^\top$ , where  $f(\Lambda) = \text{diag}(f(\lambda_1), \dots, f(\lambda_n))$ . Then, the Fréchet derivative of  $f$  at  $X$  in the direction  $E \in \mathbb{S}^n$  satisfies*

$$Df(X)[E] = U \left( f^{[1]}(\Lambda) \circ (U^\top E U) \right) U^\top, \quad (4)$$

where  $\circ$  denotes the Hadamard product and the matrix of divided differences  $f^{[1]}$  is defined by

$$f_{ij}^{[1]} = \begin{cases} \frac{f(\lambda_i) - f(\lambda_j)}{\lambda_i - \lambda_j}, & i \neq j, \\ f'(\lambda_i), & i = j. \end{cases}$$

## D. Derivation of Staleness-Induced Preconditioner Gap

In this section, we provide a derivation process for the following equation:

$$\begin{aligned} P_t - \hat{P}_t &= \{(L_t + \epsilon \mathbf{I}_m)^{-1/p} \otimes (R_t + \epsilon \mathbf{I}_n)^{-1/p}\} - (\hat{L}_t^{-1/p} \otimes \hat{R}_t^{-1/p}) \\ &= \hat{L}_t^{-1/p} \otimes \Delta_R(t) + \Delta_L(t) \otimes \hat{R}_t^{-1/p} + \Delta_L(t) \otimes \Delta_R(t) \end{aligned}$$

*Proof.* Note that  $(L_t + \epsilon \mathbf{I}_m)^{-1/p} = \Delta_L(t) + \hat{L}_t^{-1/p}$ , and  $(R_t + \epsilon \mathbf{I}_n)^{-1/p} = \Delta_R(t) + \hat{R}_t^{-1/p}$ . By the property of the Kronecker product (Lemma C.8), we get:

$$(\Delta_L(t) + \hat{L}_t^{-1/p}) \otimes (\Delta_R(t) + \hat{R}_t^{-1/p}) = (\Delta_L(t) \otimes \Delta_R(t)) + (\Delta_L(t) \otimes \hat{R}_t^{-1/p}) + (\hat{L}_t^{-1/p} \otimes \Delta_R(t)) + (\hat{L}_t^{-1/p} \otimes \hat{R}_t^{-1/p}).$$

Hence, we obtain the desired result. □

## E. Proof of Operator-Gap Staleness Error Bound (Lemma 5.2)

**Lemma E.1.** Under Assumption 5.1, for all  $\epsilon > 0$ ,  $p \geq 1$ , and  $1 \leq t \leq T$ , with probability at least  $1 - \delta$ :

$$\begin{aligned}\|\Delta_L(t)\|_2 &\leq \frac{2}{p\epsilon^{(p+1)/p}}(1 - \beta^f)R_{\text{SG}}^2, \\ \|\Delta_R(t)\|_2 &\leq \frac{2}{p\epsilon^{(p+1)/p}}(1 - \beta^f)R_{\text{SG}}^2,\end{aligned}$$

where

$$R_{\text{SG}} := R + K\sigma\sqrt{\log(T/\delta)},$$

with  $0 < \delta < 1$  and a universal constant  $K > 0$ .

*Proof.* **Step 1. (Good event)** Define

$$\mathcal{E} := \left\{ \max_{1 \leq t \leq T} \|G_t\|_2 \leq R_{\text{SG}} \right\}.$$

By the union bound,

$$\mathbb{P}(\mathcal{E}^c) \leq \sum_{t=1}^T \mathbb{P}(\|G_t\|_2 > R_{\text{SG}}).$$

By Assumption 5.1,

$$\begin{aligned}\|G_t\|_2 &\leq \|\mathbb{E}[G_t \mid \mathcal{F}_{t-1}]\|_2 + \|G_t - \mathbb{E}[G_t \mid \mathcal{F}_{t-1}]\|_2 \\ &\leq R + \|G_t - \mathbb{E}[G_t \mid \mathcal{F}_{t-1}]\|_2,\end{aligned}$$

so

$$\{\|G_t\|_2 > R_{\text{SG}}\} \subseteq \{\|G_t - \mathbb{E}[G_t \mid \mathcal{F}_{t-1}]\|_2 > R_{\text{SG}} - R\}.$$

Therefore, by the tail bound in Assumption 5.1,

$$\mathbb{P}(\|G_t\|_2 > R_{\text{SG}}) \leq \frac{\delta}{T},$$

hence  $\mathbb{P}(\mathcal{E}) \geq 1 - \delta$ .

**Step 2. (Bounding  $\|L_t\|_2, \|R_t\|_2$  on  $\mathcal{E}$ )** Under  $\mathcal{E}$ ,  $\|G_t\|_2 \leq R_{\text{SG}}$  for all  $t$ , hence  $\|G_t G_t^\top\|_2 \leq R_{\text{SG}}^2$ . We show  $\|L_t\|_2 \leq R_{\text{SG}}^2$  by induction (the same holds for  $R_t$ ). Since  $L_0 = \mathbf{0}_m \preceq R_{\text{SG}}^2 \mathbf{I}_m$ , the base case holds. Assume  $L_{t-1} \preceq R_{\text{SG}}^2 \mathbf{I}_m$ . Then

$$L_t = \beta L_{t-1} + (1 - \beta) G_t G_t^\top \preceq \beta R_{\text{SG}}^2 \mathbf{I}_m + (1 - \beta) R_{\text{SG}}^2 \mathbf{I}_m = R_{\text{SG}}^2 \mathbf{I}_m,$$

so  $\|L_t\|_2 \leq R_{\text{SG}}^2$ .

**Step 3. (Correct stale index and bounding  $\|L_t - \hat{L}_t\|_2$ )** Recall that the algorithm refreshes the preconditioner only every  $f$  steps. Note that the stale Kronecker statistics  $\hat{L}_t = L_{\tau(t)}$ ,  $\hat{R}_t := R_{\tau(t)}$ . For stale Shampoo,  $\mathcal{C} = C_f$ , hence  $\tau(t) = 1 + f \lfloor (t-1)/f \rfloor$ . Let  $\ell = t - \tau(t) \in \{0, 1, \dots, f-1\}$ . If  $\ell = 0$ , then  $L_t = \hat{L}_t$ , and the claim is trivial; hence, assume  $\ell \geq 1$ . Unrolling the EMA recursion from  $\tau(t)$  to  $t$  yields

$$L_t = \beta^\ell L_{\tau(t)} + (1 - \beta) \sum_{k=\tau(t)+1}^t \beta^{t-k} G_k G_k^\top,$$

so

$$\begin{aligned}\|L_t - \hat{L}_t\|_2 &\leq (1 - \beta^\ell) \|L_{\tau(t)}\|_2 + (1 - \beta) \sum_{k=\tau(t)+1}^t \beta^{t-k} \|G_k G_k^\top\|_2 \\ &\leq (1 - \beta^\ell) R_{\text{SG}}^2 + (1 - \beta) \sum_{j=0}^{\ell-1} \beta^j R_{\text{SG}}^2 = 2(1 - \beta^\ell) R_{\text{SG}}^2 \\ &\leq 2(1 - \beta^f) R_{\text{SG}}^2.\end{aligned}$$

Hence,

$$\|L_t - \hat{L}_t\|_2 \leq 2(1 - \beta^f)R_{\text{SG}}^2 \quad (5)$$

holds on  $\mathcal{E}$ . The same logic holds for  $\|R_t - \hat{R}_t\|_2$ .

**Step 4. (Lipschitz bound and Frobenius norm)** Apply Lemma C.1 to the scalar function  $f(\lambda) = \lambda^{-1/p}$  on  $[\epsilon, \epsilon + R_{\text{SG}}^2]$ . Since  $|f'(\lambda)| = \frac{1}{p}\lambda^{-(p+1)/p}$ , we have  $\sup_{\lambda \in [\epsilon, \epsilon + R_{\text{SG}}^2]} |f'(\lambda)| = \frac{1}{p\epsilon^{(p+1)/p}}$ . Thus, on  $\mathcal{E}$ :

$$\begin{aligned} \|\Delta_L(t)\|_2 &= \|(L_t + \epsilon \mathbf{I}_m)^{-1/p} - (\hat{L}_t + \epsilon \mathbf{I}_m)^{-1/p}\|_2 \\ &\leq \frac{1}{p\epsilon^{(p+1)/p}} \|L_t - \hat{L}_t\|_2 \leq \frac{2}{p\epsilon^{(p+1)/p}} (1 - \beta^f)R_{\text{SG}}^2, \end{aligned}$$

and similarly

$$\|\Delta_R(t)\|_2 \leq \frac{2}{p\epsilon^{(p+1)/p}} (1 - \beta^f)R_{\text{SG}}^2.$$

We obtain the desired result.  $\square$

## F. Proof of staleness-Induced Eigenvalue Gap and Eigenspace Rotation Bound

**Theorem F.1.** *Let the eigenvalues of  $(L_t + \epsilon \mathbf{I}_m)^{-1/p}$  and  $(R_t + \epsilon \mathbf{I}_n)^{-1/p}$  be sorted in non-increasing order, denoted  $\lambda_1(\cdot) \geq \lambda_2(\cdot) \geq \dots$  for all  $p \geq 1$ . Then, under Assumption 5.1, with probability at least  $1 - \delta$ , for all  $0 \leq t \leq T$ :*

$$\begin{aligned} |\lambda_i((L_t + \epsilon \mathbf{I}_m)^{-1/p}) - \lambda_i(\widehat{L}_t^{-1/p})| &\leq \|(L_t + \epsilon \mathbf{I}_m)^{-1/p} - \widehat{L}_t^{-1/p}\|_2 \\ &\leq \frac{2(1 - \beta^f)R_{SG}^2}{p\epsilon^{(p+1)/p}}, \end{aligned}$$

$$\begin{aligned} |\lambda_i((R_t + \epsilon \mathbf{I}_n)^{-1/p}) - \lambda_i(\widehat{R}_t^{-1/p})| &\leq \|(R_t + \epsilon \mathbf{I}_n)^{-1/p} - \widehat{R}_t^{-1/p}\|_2 \\ &\leq \frac{2(1 - \beta^f)R_{SG}^2}{p\epsilon^{(p+1)/p}}. \end{aligned}$$

*Proof.* By Lemma C.2 and Lemma C.1 (applied to  $f(\lambda) = \lambda^{-1/p}$ ), we obtain the claim.  $\square$

**Theorem F.2.** *Let  $A := (L_{\tau(t)} + \epsilon \mathbf{I}_m)^{-1/p}$  and  $\widehat{A} := (L_t + \epsilon \mathbf{I}_m)^{-1/p}$ , with eigenvalues  $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_{m-1} > \lambda_m$  and  $\hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \dots \geq \hat{\lambda}_{m-1} > \hat{\lambda}_m$  respectively. Let  $V_\ell := [v_1 \dots v_{m-1}] \in \mathbb{R}^{m \times (m-1)}$ ,  $\widehat{V}_\ell := [\hat{v}_1 \dots \hat{v}_{m-1}] \in \mathbb{R}^{m \times (m-1)}$  have orthonormal columns satisfying  $Av_j = \lambda_j v_j$  and  $\widehat{A}\hat{v}_j = \hat{\lambda}_j \hat{v}_j$  for  $1 \leq j \leq m-1$ . Then, under Assumption 5.1, with probability at least  $1 - \delta$ , for all  $0 \leq t \leq T$ :*

$$\begin{aligned} \|\sin \Theta(V_\ell, \widehat{V}_\ell)\|_F &\leq \frac{2 \min(\sqrt{m-1} \|\widehat{A} - A\|_2, \|\widehat{A} - A\|_F)}{\lambda_{m-1} - \lambda_m} \\ &\leq \frac{4\sqrt{m}(1 - \beta^f)R_{SG}^2}{p\epsilon^{(p+1)/p}(\lambda_{m-1} - \lambda_m)}. \end{aligned}$$

The case of  $A_R := (R_{\tau(t)} + \epsilon \mathbf{I}_n)^{-1/p}$  and  $\widehat{A}_R := (R_t + \epsilon \mathbf{I}_n)^{-1/p}$  is analogous.

*Proof.* Let  $\star := \widehat{A} - A$ . The first inequality follows from Lemma C.3. Next,

$$\min(\sqrt{m-1} \|\star\|_2, \|\star\|_F) \leq \sqrt{m} \|\star\|_2.$$

By Lemma C.1 (applied to  $f(\lambda) = \lambda^{-1/p}$ ),

$$\|\star\|_2 \leq \frac{2(1 - \beta^f)R_{SG}^2}{p\epsilon^{(p+1)/p}}.$$

Substituting yields the result.  $\square$



## G. Proof of Discounted Regret Bound (Theorem 5.4)

**Theorem G.1.** *Let us assume that Assumption 5.1 and Assumption 5.3 hold. Furthermore, assume  $\text{rank}(G_t) \leq r \leq \min(m, n)$  and  $\max_t \|W_t - W^*\|_F = D(W^*) =: D$  for all  $W^* \in \mathbb{R}^{m \times n}$ . Then, for all  $\epsilon > 0$ , with probability at least  $1 - \delta$ , the (discount) regret bound of stale Shampoo with  $p = 2$  is:*

$$\begin{aligned} \sum_{t=1}^T \beta^{T-t} f_t(W_t) - \sum_{t=1}^T \beta^{T-t} f_t(W^*) &\leq \frac{1}{2\eta} \left\{ \epsilon D^2 + \frac{\sqrt{R_{SG}^2 + \epsilon} D^2}{\sqrt{\epsilon}} R_{SG}^2 + \frac{D^2 (R_{SG}^2 + \epsilon)}{\beta} \right\} \\ &\quad + \frac{\eta}{1-\beta} \left\{ \frac{r(1-\beta^f) R_{SG}^4}{\epsilon^2} \right\} + \frac{\eta}{2(1-\beta)} \left\{ r m n \log \left( \epsilon + \frac{(1-\beta^T) r R_{SG}^2}{m n} \right) - r m n \log(\epsilon) \right\}, \end{aligned}$$

and stale Shampoo with  $p = 4$  is:

$$\begin{aligned} \sum_{t=1}^T \beta^{T-t} f_t(W_t) - \sum_{t=1}^T \beta^{T-t} f_t(W^*) &\leq \frac{1}{2\eta} \left\{ \sqrt{\epsilon} D^2 + \frac{\sqrt[4]{R_{SG}^2 + \epsilon} D^2}{2\epsilon^{3/4}} R_{SG}^2 + \frac{D^2 \sqrt{R_{SG}^2 + \epsilon}}{\beta} \right\} \\ &\quad + \underbrace{\frac{\eta r (1-\beta^f) R_{SG}^4}{2(1-\beta) \epsilon^{3/2}}}_{\text{Regret staleness error}} + \frac{\eta m n r \sqrt{R_{SG}^2 + \epsilon}}{1-\beta}. \end{aligned}$$

*Proof.* We proceed under the good event  $\mathcal{E}$ , which holds with probability at least  $1 - \delta$ .

**Preconditioners and phases.** For  $t \geq 1$  and  $p \geq 1$ , define

$$H_{t,p} = (L_t + \epsilon \mathbf{I}_m)^{1/p} \otimes (R_t + \epsilon \mathbf{I}_n)^{1/p}, \quad (L_0 = \mathbf{0}_m, R_0 = \mathbf{0}_n).$$

Assume for simplicity that  $T = \mathbf{f}K$  (otherwise, take  $K = \lceil T/\mathbf{f} \rceil$  and the last phase is shorter; the same bounds apply). Define phase indices  $T_k = (k-1)\mathbf{f} + 1$  and phases  $t \in \{T_k, \dots, T_{k+1} - 1\}$ . Define for each phase  $k$ :

$$L_k := L_{T_k}, \quad R_k := R_{T_k}, \quad H_{k,p} := (L_k + \epsilon \mathbf{I}_m)^{1/p} \otimes (R_k + \epsilon \mathbf{I}_n)^{1/p}.$$

For  $t$  in phase  $k$ , stale Shampoo uses the fixed preconditioner  $H_{k,p}$ .

**Step 1.** Let  $w_t = \text{vec}(W_t)$ ,  $g_t = \text{vec}(G_t)$ . Multiplying by  $(1-\beta)$ , applying convexity and Lemma C.8, we have:

$$(1-\beta) \sum_{t=1}^T \beta^{T-t} (f_t(W_t) - f_t(W^*)) \leq (1-\beta) \sum_{t=1}^T \beta^{T-t} \langle G_t, W_t - W^* \rangle = (1-\beta) \sum_{t=1}^T \beta^{T-t} g_t^\top (w_t - w^*).$$

By Lemma 3 of Gupta et al. (2017),

$$(1-\beta) \sum_{t=1}^T \beta^{T-t} g_t^\top (w_t - w^*) \leq \underbrace{\frac{1}{2\eta} (1-\beta) \sum_{t=1}^T \beta^{T-t} \Delta_t(w^*)}_{(a)} + \underbrace{\frac{\eta}{2} (1-\beta) \sum_{t=1}^T \beta^{T-t} \|g_t\|_{H_{t,p}}^2}_{(b)},$$

where  $\Delta_t(x^*) := \|x_t - x^*\|_{H_{t,p}}^2 - \|x_{t+1} - x^*\|_{H_{t,p}}^2$ .

**Step 2: Bounding (a).** Since  $H_{t,p} = H_{k(t),p}$  within each phase,

$$(a) = (1-\beta) \sum_{k=1}^K \sum_{t=T_k}^{T_{k+1}-1} \beta^{T-t} \left( \|w_t - w^*\|_{H_{k,p}}^2 - \|w_{t+1} - w^*\|_{H_{k,p}}^2 \right).$$

Let  $k(t)$  be the phase index containing  $t$  and define  $A_t := \|w_t - w^*\|_{H_{k(t),p}}^2$  and  $\tilde{A}_{t+1} := \|w_{t+1} - w^*\|_{H_{k(t),p}}^2$ . Then

$$(a) = (1-\beta) \sum_{t=1}^T \beta^{T-t} (A_t - \tilde{A}_{t+1}).$$

Add and subtract  $A_{t+1}$ :

$$A_t - \tilde{A}_{t+1} = (A_t - A_{t+1}) + (A_{t+1} - \tilde{A}_{t+1}),$$

hence

$$(a) = (1 - \beta) \sum_{t=1}^T \beta^{T-t} (A_t - A_{t+1}) + (1 - \beta) \sum_{t=1}^T \beta^{T-t} (A_{t+1} - \tilde{A}_{t+1}).$$

(i) **Discounted telescoping term.** For any sequence  $\{A_t\}$ ,

$$(1 - \beta) \sum_{t=1}^T \beta^{T-t} (A_t - A_{t+1}) = (1 - \beta) [\beta^{T-1} A_1 - \beta^{-1} A_{T+1}] + (1 - \beta)^2 \sum_{t=2}^T \beta^{T-t} A_t.$$

Dropping the negative term  $-\beta^{-1} A_{T+1}$  and using  $\|w_t - w^*\|_2 \leq D$  and by Lemma C.8, we get  $\|H_{k(t),p}\|_2 \leq (R_{\text{SG}}^2 + \epsilon)^{2/p}$  on  $\mathcal{E}$ , hence:

$$\begin{aligned} (1 - \beta) \sum_{t=1}^T \beta^{T-t} (A_t - A_{t+1}) &\leq (1 - \beta) \beta^{T-1} \|w_1 - w^*\|_{H_{1,p}}^2 + (1 - \beta)^2 \sum_{t=2}^T \beta^{T-t} \|w_t - w^*\|_{H_{k(t),p}}^2 \\ &\leq (1 - \beta) \epsilon^{2/p} D^2 + (1 - \beta)^2 D^2 (R_{\text{SG}}^2 + \epsilon)^{2/p} \sum_{t=2}^T \beta^{T-t} \\ &\leq (1 - \beta) \epsilon^{2/p} D^2 + (1 - \beta) \frac{1 - \beta^T}{\beta} D^2 (R_{\text{SG}}^2 + \epsilon)^{2/p}. \end{aligned}$$

(ii) **Phase-boundary correction term.** Note  $A_{t+1} - \tilde{A}_{t+1} = 0$  unless  $t = T_k - 1$  for some  $k \geq 2$ , in which case

$$A_{T_k} - \tilde{A}_{T_k} = \|w_{T_k} - w^*\|_{H_{k,p}}^2 - \|w_{T_k} - w^*\|_{H_{k-1,p}}^2.$$

Therefore,

$$\begin{aligned} (1 - \beta) \sum_{t=1}^T \beta^{T-t} (A_{t+1} - \tilde{A}_{t+1}) &= (1 - \beta) \sum_{k=2}^K \beta^{T-(T_k-1)} \left( \|w_{T_k} - w^*\|_{H_{k,p}}^2 - \|w_{T_k} - w^*\|_{H_{k-1,p}}^2 \right) \\ &\leq (1 - \beta) \sum_{k=2}^K \beta^{T-T_k} D^2 \|H_{k,p} - H_{k-1,p}\|_2, \end{aligned}$$

where we used  $\beta^{T-(T_k-1)} = \beta \beta^{T-T_k} \leq \beta^{T-T_k}$  since  $\beta \in (0, 1)$ . On  $\mathcal{E}$ , for  $k \geq 2$ , we have:

$$\|L_k - L_{k-1}\|_2 \leq 2(1 - \beta^f) R_{\text{SG}}^2, \quad \|R_k - R_{k-1}\|_2 \leq 2(1 - \beta^f) R_{\text{SG}}^2.$$

via Lemma 5.2.

Using Lemma C.1, Lemma C.8 and the inequality

$$\|A \otimes B - C \otimes D\|_2 \leq \|A - C\|_2 \|B\|_2 + \|C\|_2 \|B - D\|_2,$$

we obtain

$$\begin{aligned} \|H_{k,p} - H_{k-1,p}\|_2 &= \|(L_k + \epsilon \mathbf{I}_m)^{1/p} \otimes (R_k + \epsilon \mathbf{I}_n)^{1/p} - (L_{k-1} + \epsilon \mathbf{I}_m)^{1/p} \otimes (R_{k-1} + \epsilon \mathbf{I}_n)^{1/p}\|_2 \\ &\leq \|(L_k + \epsilon \mathbf{I}_m)^{1/p} - (L_{k-1} + \epsilon \mathbf{I}_m)^{1/p}\|_2 \|(R_k + \epsilon \mathbf{I}_n)^{1/p}\|_2 \\ &\quad + \|(L_{k-1} + \epsilon \mathbf{I}_m)^{1/p}\|_2 \|(R_k + \epsilon \mathbf{I}_n)^{1/p} - (R_{k-1} + \epsilon \mathbf{I}_n)^{1/p}\|_2 \\ &\leq \frac{1}{p} \epsilon^{(1-p)/p} \|L_k - L_{k-1}\|_2 (R_{\text{SG}}^2 + \epsilon)^{1/p} + \frac{1}{p} \epsilon^{(1-p)/p} \|R_k - R_{k-1}\|_2 (R_{\text{SG}}^2 + \epsilon)^{1/p} \\ &\leq \frac{4(1 - \beta^f)}{p} \epsilon^{(1-p)/p} (R_{\text{SG}}^2 + \epsilon)^{1/p} R_{\text{SG}}^2. \end{aligned}$$

Hence

$$(1 - \beta) \sum_{t=1}^T \beta^{T-t} (A_{t+1} - \tilde{A}_{t+1}) \leq (1 - \beta) D^2 \frac{4(1 - \beta^f)}{p} \epsilon^{(1-p)/p} (R_{\text{SG}}^2 + \epsilon)^{1/p} R_{\text{SG}}^2 \sum_{k=2}^K \beta^{T-T_k}.$$

Finally,

$$\sum_{k=2}^K \beta^{T-T_k} = \sum_{k=2}^K \beta^{Kf - ((k-1)f+1)} = \beta^{-1} \sum_{k=2}^K (\beta^f)^{K-k+1} \leq \frac{\beta^{f-1}}{1 - \beta^f}.$$

Therefore,

$$(1 - \beta) \sum_{t=1}^T \beta^{T-t} (A_{t+1} - \tilde{A}_{t+1}) \leq \beta^{f-1} \frac{4(1 - \beta)}{p} \epsilon^{(1-p)/p} (R_{\text{SG}}^2 + \epsilon)^{1/p} D^2 R_{\text{SG}}^2.$$

**Conclusion for (a).** Combining (i) and (ii):

$$(a) \leq (1 - \beta) \epsilon^{2/p} D^2 + \beta^{f-1} \frac{4(1 - \beta)}{p} \epsilon^{(1-p)/p} (R_{\text{SG}}^2 + \epsilon)^{1/p} D^2 R_{\text{SG}}^2 + (1 - \beta) \frac{1 - \beta^T}{\beta} D^2 (R_{\text{SG}}^2 + \epsilon)^{2/p}.$$

**Step 3: Bounding (b).** Write

$$(b) = (1 - \beta) \left\{ \sum_{k=1}^K \sum_{t=T_k}^{T_{k+1}-1} \beta^{T-t} \left( \|g_t\|_{H_{k,p}}^{*2} - \|g_t\|_{H_{t,p}}^{*2} \right) + \sum_{t=1}^T \beta^{T-t} \|g_t\|_{H_{t,p}}^{*2} \right\}. \quad (6)$$

We consider  $p = 2$  and  $p = 4$ .

### 1. Stale Shampoo with $p = 2$ .

**(b1) Stale–fresh gap term.** For  $t$  in phase  $k$ ,

$$\begin{aligned} \|g_t\|_{H_{k,2}}^{*2} - \|g_t\|_{H_{t,2}}^{*2} &= g_t^\top \left( (H_{k,2})^{-1} - (H_{t,2})^{-1} \right) g_t \\ &\leq \|g_t\|_2^2 \|(H_{k,2})^{-1} - (H_{t,2})^{-1}\|_2 \leq r R_{\text{SG}}^2 \|(H_{k,2})^{-1} - (H_{t,2})^{-1}\|_2, \end{aligned}$$

where  $\|g_t\|_2^2 = \|G_t\|_F^2 \leq r \|G_t\|_2^2 \leq r R_{\text{SG}}^2$  on  $\mathcal{E}$ .

Let  $\hat{L}_t = L_k$  and  $\hat{R}_t = R_k$  for  $t$  in phase  $k$ . Then by Lemma 5.2 with  $p = 2$ :

$$\|(L_t + \epsilon \mathbf{I}_m)^{-1/2} - (L_k + \epsilon \mathbf{I}_m)^{-1/2}\|_2 \leq \frac{2(1 - \beta^f) R_{\text{SG}}^2}{2 \epsilon^{3/2}} = \frac{(1 - \beta^f) R_{\text{SG}}^2}{\epsilon^{3/2}},$$

and similarly for  $R$ . Using the Kronecker difference inequality and  $\|(R_k + \epsilon \mathbf{I}_n)^{-1/2}\|_2 \leq \epsilon^{-1/2}$ , we obtain

$$\|(H_{k,2})^{-1} - (H_{t,2})^{-1}\|_2 \leq \frac{2(1 - \beta^f) R_{\text{SG}}^2}{\epsilon^2}.$$

Therefore,

$$\begin{aligned} (1 - \beta) \sum_{k=1}^K \sum_{t=T_k}^{T_{k+1}-1} \beta^{T-t} \left( \|g_t\|_{H_{k,2}}^{*2} - \|g_t\|_{H_{t,2}}^{*2} \right) &\leq (1 - \beta) \sum_{t=1}^T \beta^{T-t} r R_{\text{SG}}^2 \cdot \frac{2(1 - \beta^f) R_{\text{SG}}^2}{\epsilon^2} \\ &= \frac{2r(1 - \beta^f)(1 - \beta^T) R_{\text{SG}}^4}{\epsilon^2}. \end{aligned}$$

**(b2) Fresh term via FTL–BTL + logdet.** We apply Lemma C.4 with

$$\Phi(H) = \log \det(H) + \epsilon \text{Tr}(H^{-1}), \quad S_t := (1 - \beta) \sum_{s=1}^t \beta^{T-s} g_s g_s^\top.$$

Define

$$\hat{H}_t \in \arg \min_{H \succ 0} \left\{ \langle S_t, H^{-1} \rangle_F + \log \det(H) + \epsilon \operatorname{Tr}(H^{-1}) \right\}.$$

Using standard matrix calculus,

$$\hat{H}_t = \epsilon \mathbf{I}_{mn} + S_t.$$

By Lemma C.4:

$$(1 - \beta) \sum_{t=1}^T \beta^{T-t} \|g_t\|_{\hat{H}_t}^{\star 2} \leq (1 - \beta) \sum_{t=1}^T \beta^{T-t} \|g_t\|_{\hat{H}_T}^{\star 2} + \Phi(\hat{H}_T) - \Phi(\hat{H}_0).$$

Moreover,

$$(1 - \beta) \sum_{t=1}^T \beta^{T-t} \|g_t\|_{\hat{H}_T}^{\star 2} + \epsilon \operatorname{Tr}(\hat{H}_T^{-1}) = \langle S_T, \hat{H}_T^{-1} \rangle_F + \epsilon \operatorname{Tr}(\hat{H}_T^{-1}) = \langle \epsilon \mathbf{I}_{mn} + S_T, \hat{H}_T^{-1} \rangle_F = \operatorname{Tr}(\hat{H}_T \hat{H}_T^{-1}) = mn.$$

Since  $\hat{H}_0 = \epsilon \mathbf{I}_{mn}$ , we have  $\log \det(\hat{H}_0) = mn \log \epsilon$  and  $\epsilon \operatorname{Tr}(\hat{H}_0^{-1}) = mn$ . Therefore,

$$(1 - \beta) \sum_{t=1}^T \beta^{T-t} \|g_t\|_{\hat{H}_t}^{\star 2} \leq \log \det(\hat{H}_T) - mn \log \epsilon.$$

**Relating  $\hat{H}_t$  and  $H_{t,2}$ .** For  $t \leq T$ , by Lemma C.7 applied to  $\{G_s\}_{s \leq t}$ ,

$$\epsilon \mathbf{I}_{mn} + \frac{1}{r} S_t \preceq A_t^{1/2} \otimes B_t^{1/2}, \quad A_t := \epsilon \mathbf{I}_m + (1 - \beta) \sum_{s=1}^t \beta^{T-s} G_s G_s^\top, \quad B_t := \epsilon \mathbf{I}_n + (1 - \beta) \sum_{s=1}^t \beta^{T-s} G_s^\top G_s.$$

Since  $r \geq 1$ ,

$$\hat{H}_t = \epsilon \mathbf{I}_{mn} + S_t \preceq r \left( \epsilon \mathbf{I}_{mn} + \frac{1}{r} S_t \right) \preceq r (A_t^{1/2} \otimes B_t^{1/2}).$$

Also, using  $\beta^{T-s} = \beta^{T-t} \beta^{t-s}$  and  $\beta^{T-t} \leq 1$ :

$$A_t = \epsilon \mathbf{I}_m + \beta^{T-t} L_t \preceq \epsilon \mathbf{I}_m + L_t, \quad B_t = \epsilon \mathbf{I}_n + \beta^{T-t} R_t \preceq \epsilon \mathbf{I}_n + R_t.$$

By Lemma C.9 (square-root monotonicity) and Lemma C.8,

$$A_t^{1/2} \otimes B_t^{1/2} \preceq (\epsilon \mathbf{I}_m + L_t)^{1/2} \otimes (\epsilon \mathbf{I}_n + R_t)^{1/2} =: H_{t,2}.$$

Thus  $\hat{H}_t \preceq r H_{t,2}$ , hence  $H_{t,2}^{-1} \preceq r \hat{H}_t^{-1}$  and

$$\|g_t\|_{H_{t,2}}^{\star 2} \leq r \|g_t\|_{\hat{H}_t}^{\star 2}.$$

Therefore,

$$(1 - \beta) \sum_{t=1}^T \beta^{T-t} \|g_t\|_{H_{t,2}}^{\star 2} \leq r (1 - \beta) \sum_{t=1}^T \beta^{T-t} \|g_t\|_{\hat{H}_t}^{\star 2} \leq r \log \det(\hat{H}_T) - r mn \log \epsilon.$$

**Bounding  $\log \det(\hat{H}_T)$  via trace.** Since  $\hat{H}_T = \epsilon \mathbf{I}_{mn} + S_T$ ,

$$\begin{aligned} \operatorname{Tr}(\hat{H}_T) &= mn\epsilon + (1 - \beta) \sum_{s=1}^T \beta^{T-s} \operatorname{Tr}(g_s g_s^\top) = mn\epsilon + (1 - \beta) \sum_{s=1}^T \beta^{T-s} \|g_s\|_2^2 \\ &\leq mn\epsilon + (1 - \beta) \sum_{s=1}^T \beta^{T-s} r R_{\text{SG}}^2 = mn\epsilon + (1 - \beta^T) r R_{\text{SG}}^2, \end{aligned}$$

where we used  $\|g_s\|_2^2 = \|G_s\|_F^2 \leq r\|G_s\|_2^2 \leq rR_{\text{SG}}^2$  on  $\mathcal{E}$ . By Lemma C.10,

$$\log \det(\hat{H}_T) \leq mn \log \left( \frac{\text{Tr}(\hat{H}_T)}{mn} \right) \leq mn \log \left( \epsilon + \frac{(1 - \beta^T)rR_{\text{SG}}^2}{mn} \right).$$

Hence

$$(1 - \beta) \sum_{t=1}^T \beta^{T-t} \|g_t\|_{H_{t,2}}^{\star 2} \leq rmn \log \left( \epsilon + \frac{(1 - \beta^T)rR_{\text{SG}}^2}{mn} \right) - rmn \log \epsilon.$$

**Combine (a) and (b) for  $p = 2$ .** Plug the bounds for (a) and (b) into  $\frac{1}{2\eta}(a) + \frac{\eta}{2}(b)$  and use  $1 - \beta^T \leq 1$  to simplify; finally divide by  $(1 - \beta)$  to obtain the stated regret bound.

## 2. Stale Shampoo with $p = 4$ .

**(b1) Stale–fresh gap term.** For  $t$  in phase  $k$ :

$$\|g_t\|_{H_{k,4}}^{\star 2} - \|g_t\|_{H_{t,4}}^{\star 2} \leq \|g_t\|_2^2 \|(H_{k,4})^{-1} - (H_{t,4})^{-1}\|_2 \leq rR_{\text{SG}}^2 \|(H_{k,4})^{-1} - (H_{t,4})^{-1}\|_2.$$

From Lemma 5.2 with  $p = 4$ ,

$$\|(L_t + \epsilon I_m)^{-1/4} - (L_k + \epsilon I_m)^{-1/4}\|_2 \leq \frac{2(1 - \beta^f)R_{\text{SG}}^2}{4\epsilon^{5/4}} = \frac{(1 - \beta^f)R_{\text{SG}}^2}{2\epsilon^{5/4}},$$

and similarly for  $R$ . Using  $\|(R_k + \epsilon I_n)^{-1/4}\|_2 \leq \epsilon^{-1/4}$  and the Kronecker difference inequality,

$$\|(H_{k,4})^{-1} - (H_{t,4})^{-1}\|_2 \leq \frac{(1 - \beta^f)R_{\text{SG}}^2}{\epsilon^{3/2}}.$$

Therefore,

$$(1 - \beta) \sum_{k=1}^K \sum_{t=T_k}^{T_{k+1}-1} \beta^{T-t} \left( \|g_t\|_{H_{k,4}}^{\star 2} - \|g_t\|_{H_{t,4}}^{\star 2} \right) \leq \frac{r(1 - \beta^f)(1 - \beta^T)R_{\text{SG}}^4}{\epsilon^{3/2}}.$$

**(b2) Fresh term via FTL–BTL (trace potential).** Apply Lemma C.4 with

$$\Phi(H) = \text{Tr}(H) + \epsilon r \text{Tr}(H^{-1}), \quad S_t = (1 - \beta) \sum_{s=1}^t \beta^{T-s} g_s g_s^\top,$$

and define

$$\hat{H}_t \in \arg \min_{H \succ 0} \left\{ \langle S_t, H^{-1} \rangle_F + \text{Tr}(H) + \epsilon r \text{Tr}(H^{-1}) \right\}.$$

First-order optimality yields

$$\hat{H}_t = (\epsilon r \mathbf{I}_{mn} + S_t)^{1/2}.$$

Lemma C.4 gives

$$(1 - \beta) \sum_{t=1}^T \beta^{T-t} \|g_t\|_{\hat{H}_t}^{\star 2} \leq (1 - \beta) \sum_{t=1}^T \beta^{T-t} \|g_t\|_{\hat{H}_T}^{\star 2} + \Phi(\hat{H}_T) - \Phi(\hat{H}_0).$$

Using  $\hat{H}_0 = (\epsilon r)^{1/2} \mathbf{I}_{mn}$  and  $\hat{H}_T^2 = \epsilon r \mathbf{I}_{mn} + S_T$ ,

$$(1 - \beta) \sum_{t=1}^T \beta^{T-t} \|g_t\|_{\hat{H}_T}^{\star 2} + \epsilon r \text{Tr}(\hat{H}_T^{-1}) = \langle \epsilon r \mathbf{I}_{mn} + S_T, \hat{H}_T^{-1} \rangle_F = \text{Tr}(\hat{H}_T^2 \hat{H}_T^{-1}) = \text{Tr}(\hat{H}_T).$$

Hence

$$(1 - \beta) \sum_{t=1}^T \beta^{T-t} \|g_t\|_{\hat{H}_t}^{\star 2} \leq 2 \text{Tr}(\hat{H}_T).$$

**Relating  $\hat{H}_t$  and  $H_{t,4}$ .** By Lemma C.7 applied to  $\{G_s\}_{s \leq t}$ ,

$$\epsilon \mathbf{I}_{mn} + \frac{1}{r} S_t \preceq A_t^{1/2} \otimes B_t^{1/2} \Rightarrow \epsilon r \mathbf{I}_{mn} + S_t \preceq r(A_t^{1/2} \otimes B_t^{1/2}),$$

with the same  $A_t, B_t$  as above. Taking square roots and using Lemma C.9 and Lemma C.8,

$$\hat{H}_t = (\epsilon r \mathbf{I}_{mn} + S_t)^{1/2} \preceq \sqrt{r} (A_t^{1/2} \otimes B_t^{1/2})^{1/2} = \sqrt{r} (A_t^{1/4} \otimes B_t^{1/4}).$$

Since  $A_t \preceq \epsilon \mathbf{I}_m + L_t$  and  $B_t \preceq \epsilon \mathbf{I}_n + R_t$ ,

$$A_t^{1/4} \otimes B_t^{1/4} \preceq (\epsilon \mathbf{I}_m + L_t)^{1/4} \otimes (\epsilon \mathbf{I}_n + R_t)^{1/4} =: H_{t,4}.$$

Therefore  $\hat{H}_t \preceq \sqrt{r} H_{t,4}$ , so  $H_{t,4}^{-1} \preceq \sqrt{r} \hat{H}_t^{-1}$  and

$$\|g_t\|_{H_{t,4}}^{*2} \leq \sqrt{r} \|g_t\|_{\hat{H}_t}^{*2}.$$

Thus

$$(1 - \beta) \sum_{t=1}^T \beta^{T-t} \|g_t\|_{H_{t,4}}^{*2} \leq \sqrt{r} (1 - \beta) \sum_{t=1}^T \beta^{T-t} \|g_t\|_{\hat{H}_t}^{*2} \leq 2\sqrt{r} \text{Tr}(\hat{H}_T) \leq 2r \text{Tr}(H_{T,4}),$$

where the last inequality uses  $\hat{H}_T \preceq \sqrt{r} H_{T,4}$  and monotonicity of trace on PSD.

Finally,

$$\text{Tr}(H_{T,4}) = \text{Tr}((L_T + \epsilon \mathbf{I}_m)^{1/4}) \text{Tr}((R_T + \epsilon \mathbf{I}_n)^{1/4}) \leq m(R_{\text{SG}}^2 + \epsilon)^{1/4} \cdot n(R_{\text{SG}}^2 + \epsilon)^{1/4} = mn\sqrt{R_{\text{SG}}^2 + \epsilon},$$

so

$$(1 - \beta) \sum_{t=1}^T \beta^{T-t} \|g_t\|_{H_{t,4}}^{*2} \leq 2mnr\sqrt{R_{\text{SG}}^2 + \epsilon}.$$

**Combine (a) and (b) for  $p = 4$ .** Plug the bounds for (a) and (b) into  $\frac{1}{2\eta}(a) + \frac{\eta}{2}(b)$ , use  $1 - \beta^T \leq 1$ , and finally divide by  $(1 - \beta)$  to obtain the stated bound.

□



## H. Proof of Approximation of Relative Operator Error

**Theorem H.1** (Relative operator error for inverse  $p$ -th Root). *Let  $A \in \mathbb{S}_+^n$  be a positive definite matrix and  $E \in \mathbb{S}^n$  be a perturbation matrix. For any  $p > 0$ , the relative error of the inverse  $p$ -th root is bounded by:*

$$\frac{\|(A + E)^{-1/p} - A^{-1/p}\|_F}{\|A^{-1/p}\|_F} \leq \frac{\|(A + E)^{-1/p} - A^{-1/p}\|_F}{\|A^{-1/p}\|_2} \approx \frac{\|\Delta_1\|_F}{\|A^{-1/p}\|_2} \leq \frac{1}{p} \|A^{-1/2} E A^{-1/2}\|_F,$$

where  $\approx$  denotes the first-order approximation.

*Proof.* We want to show how much  $(A + E)^{-1/p}$  differs from  $A^{-1/p}$  when matrix  $A$  undergoes perturbation  $E$  to become  $A + E$ . Let  $A = U\Lambda U^T$  be the eigendecomposition of  $A$ , where  $\Lambda = \text{diag}(\mu_1, \dots, \mu_n)$  has eigenvalues  $\mu_i > 0$ . Given  $F(A) = A^{-1/p}$  and  $\Delta = F(A + E) - F(A)$ , we have:

$$F(A + E) = F(A) + \Delta = F(A) + \underbrace{D_F(A)[E]}_{\Delta_1} + R_2,$$

where  $\Delta_1$  is the first-order term and  $R_2$  is the remainder term of the Fréchet derivative. We rotate the perturbation  $E$  and  $\Delta_1$  into the eigenbasis:

$$\hat{E} = U^T E U, \quad \hat{\Delta}_1 = U^T \Delta_1 U.$$

Since the Frobenius norm is orthogonally invariant, we have  $\|\Delta_1\|_F = \|\hat{\Delta}_1\|_F$  and  $\|E\|_F = \|\hat{E}\|_F$ . The first-order term  $\Delta_1$  corresponds to the Fréchet derivative of the function  $f(t) = t^{-1/p}$  applied to  $E$ . In the eigenbasis, the entries of  $\hat{\Delta}_1$  are given by the divided differences (also known as the Daleckii-Krein Theorem; Lemma C.11):

$$[\hat{\Delta}_1]_{ij} = \frac{\mu_i^{-1/p} - \mu_j^{-1/p}}{\mu_i - \mu_j} [\hat{E}]_{ij}.$$

(Note: If  $\mu_i = \mu_j$ , the fraction converges to the derivative  $f'(\mu_i)$ ). Now, recall the definition of the whitened perturbation  $\tilde{E} = A^{-1/2} E A^{-1/2}$ . In the rotated basis, its entries  $[\hat{\tilde{E}}]_{ij}$  satisfy:

$$[\hat{\tilde{E}}]_{ij} = [U^T A^{-1/2} E A^{-1/2} U]_{ij} = \frac{[\hat{E}]_{ij}}{\sqrt{\mu_i \mu_j}} \implies [\hat{E}]_{ij} = \sqrt{\mu_i \mu_j} [\hat{\tilde{E}}]_{ij}.$$

Substituting this relation into the expression for  $[\hat{\Delta}_1]_{ij}$ , we obtain:

$$[\hat{\Delta}_1]_{ij} = \underbrace{\left( \frac{\mu_i^{-1/p} - \mu_j^{-1/p}}{\mu_i - \mu_j} \sqrt{\mu_i \mu_j} \right)}_{C_{ij}} [\hat{\tilde{E}}]_{ij}.$$

We now bound the coefficient  $C_{ij}$ . Let  $q = 1/p > 0$  denote. Without loss of generality, assume that  $x = \mu_i \leq \mu_j = rx$ , with  $r \geq 1$ .

$$|C_{ij}| = x^{-q} \frac{1 - r^{-q}}{r - 1} \sqrt{r}.$$

If  $H(r) = \frac{(1-r^{-q})\sqrt{r}}{r-1} \leq q$ , then  $\max_{i,j} |C_{i,j}| \leq 1/p(\min_k \mu_k)^{-1/p} = 1/p\|A^{-1/p}\|_2$ . Let us define

$$\begin{aligned} G(r) &= q(r-1) - H(r)(r-1) \\ &= q(r-1) - \sqrt{r} + r^{\frac{1}{2}-q}. \end{aligned}$$

Then, by simple calculation, we have:  $G(1) = 0$ ,  $G'(1) = 0$  and

$$\begin{aligned} G'(r) &= q - \frac{1}{2}r^{-1/2} + \left(\frac{1}{2} - q\right)r^{-\frac{1}{2}-q} \\ G''(r) &= r^{-\frac{3}{2}-q} \left(\frac{1}{4}r^q - \frac{1}{4} + q^2\right). \end{aligned}$$

Hence, we have  $G''(r) > 0$  for all  $r \geq 1$ , and this gives a result  $G(r) \geq 0$ . Lastly, by L'Hôpital's rule, we have  $\lim_{r \rightarrow 1} H(r) = q$ . Therefore, we have  $H(r) \leq q$ , and we obtain the following result:

$$\max_{i,j} |C_{i,j}| \leq \frac{1}{p} (\min_k \mu_k)^{-1/p} = \frac{1}{p} \|A^{-1/p}\|_2.$$

Finally, we compute the Frobenius norm of  $\Delta_1$ :

$$\begin{aligned} \|\Delta_1\|_F^2 &= \sum_{i,j} \left| [\hat{\Delta}_1]_{ij} \right|^2 = \sum_{i,j} \left| C_{ij} [\hat{E}]_{ij} \right|^2 \\ &\leq \left( \max_{i,j} |C_{ij}| \right)^2 \sum_{i,j} \left| [\hat{E}]_{ij} \right|^2 \\ &= \left( \frac{1}{p} \|A^{-1/p}\|_2 \right)^2 \|\hat{E}\|_F^2. \end{aligned}$$

Taking the square root and using  $\|\hat{E}\|_F = \|\tilde{E}\|_F = \|A^{-1/2}EA^{-1/2}\|_F$ , we obtain:

$$\|\Delta_1\|_F \leq \frac{1}{p} \|A^{-1/p}\|_2 \cdot \|A^{-1/2}EA^{-1/2}\|_F \leq \frac{1}{p} \|A^{-1/p}\|_F \cdot \|A^{-1/2}EA^{-1/2}\|_F.$$

Hence, we obtain the desired results. □

## I. Further Discussion of the Decision Criterion of Eschenhagen et al. (2026) and Ours

This section clarifies a key distinction between the quantity controlled by the criterion of ? and the quantity that directly governs the stability of Shampoo-style updates. Their criterion measures how well a stale eigenbasis diagonalizes the *current Kronecker statistic*, or equivalently, how accurately the matrix itself is approximated in that stale basis. By contrast, the Shampoo update depends on the *inverse-root operator*

$$F_\epsilon(X) := (X + \epsilon \mathbf{I})^{-1/p},$$

and hence the relevant quantity is not the matrix approximation error itself, but the induced *relative inverse-root operator error*. These two notions need not agree. In particular, a basis may look nearly diagonal under a residual criterion while still inducing a substantial error after the nonlinear inverse-root map is applied.

To make this distinction precise, we first provide a simple counterexample showing that the diagonalization residual may vanish while the inverse-root operator error remains  $\Theta(1)$ . We then explain the mechanism via first-order perturbation theory for matrix functions, which reveals that (i) eigenvalue drift and (ii) eigenspace mixing enter the inverse-root map in fundamentally different, anisotropic ways. Finally, we revisit the “warm-started QR” interpretation of ? and argue that, under their trigger policy, the QR refinement is often invoked precisely in a regime that is operationally closer to a *cold start* than to a genuinely local warm-start refinement.

**What their criterion controls, and what ours controls.** Let  $\hat{Q}$  be a stale eigenbasis and define the stale-basis diagonal approximation

$$\hat{L}_t := \hat{Q} \text{diag}(\hat{Q}^\top L_t \hat{Q}) \hat{Q}^\top.$$

The criterion of ? controls the relative approximation error

$$\frac{\|L_t - \hat{L}_t\|_F}{\|L_t\|_F} = \frac{\|\hat{Q}^\top L_t \hat{Q} - \text{diag}(\hat{Q}^\top L_t \hat{Q})\|_F}{\|\hat{Q}^\top L_t \hat{Q}\|_F},$$

that is, the off-diagonal mass of the current statistic in the stale basis. This is a meaningful quantity for assessing whether  $\hat{Q}$  approximately diagonalizes  $L_t$ . However, the Shampoo update is driven not by  $L_t$  itself, but by  $(L_t + \epsilon \mathbf{I}_m)^{-1/p}$ . Therefore, from the viewpoint of update accuracy and numerical stability, the more relevant quantity is the *relative inverse-root operator error*

$$\frac{\|F_\epsilon(L_t) - F_\epsilon(\hat{L}_t)\|_F}{\|F_\epsilon(\hat{L}_t)\|_F}.$$

Our proxy is designed to approximate this latter quantity directly, rather than the former.

**A counterexample: small diagonalization residual but large inverse-root operator error.** Consider the  $2 \times 2$  SPD matrix

$$L_\delta = \begin{bmatrix} \delta & c\sqrt{\delta} \\ c\sqrt{\delta} & 1 \end{bmatrix}, \quad 0 < c < 1,$$

with  $\delta > 0$  sufficiently small. The condition  $0 < c < 1$  ensures  $L_\delta \succ 0$  for all sufficiently small  $\delta$ . Suppose the stale eigenbasis is the standard basis, i.e.  $\hat{Q} = \mathbf{I}$ . Then the diagonalization residual used by ? is

$$\frac{\|L_\delta - \text{diag}(L_\delta)\|_F}{\|\text{diag}(L_\delta)\|_F} = \frac{\sqrt{2} c \sqrt{\delta}}{\sqrt{\delta^2 + 1}} \xrightarrow{\delta \rightarrow 0} 0.$$

Hence, from the viewpoint of diagonalization residual, the stale basis appears asymptotically “accurate.”

However, the inverse-root operator error behaves very differently. For  $p = 2$ , compare the exact inverse root with the diagonal approximation in the same basis:

$$\frac{\|L_\delta^{-1/2} - \text{diag}(L_\delta)^{-1/2}\|_F}{\|\text{diag}(L_\delta)^{-1/2}\|_F} \xrightarrow{\delta \rightarrow 0} \frac{1}{\sqrt{1 - c^2}} - 1.$$

In particular, this limit is strictly positive for every fixed  $c \in (0, 1)$ . Thus, although the off-diagonal entry  $c\sqrt{\delta}$  vanishes, the relative inverse-root error does *not* vanish. Equivalently, the diagonalization residual can be arbitrarily small while the operator error relevant to the Shampoo update remains  $\Theta(1)$ .

The reason is transparent from the smallest eigenvalue. A direct calculation shows

$$\lambda_{\min}(L_\delta) = \frac{\delta + 1 - \sqrt{(1 - \delta)^2 + 4c^2\delta}}{2} = (1 - c^2)\delta + O(\delta^2), \quad \delta \downarrow 0.$$

Hence the low-spectrum direction becomes increasingly ill-conditioned, and the inverse-root map amplifies perturbations in precisely that direction. The diagonalization residual does not capture this effect.

**Why this happens: the inverse-root map is anisotropic.** The above phenomenon is not peculiar to  $2 \times 2$  matrices; it reflects the general anisotropy of matrix inverse-root maps. Let

$$X = Q\Lambda Q^\top, \quad \Lambda = \text{diag}(\lambda_1, \dots, \lambda_n),$$

and let  $\Delta X$  be a symmetric perturbation. By the Daleckiĭ–Kreĭn formula (Lemma C.11) for matrix functions, the first-order variation is

$$F_\epsilon(X + \Delta X) - F_\epsilon(X) \approx DF_\epsilon(X)[\Delta X] = Q(G \odot (Q^\top \Delta X Q))Q^\top,$$

where  $\odot$  is the Hadamard product and  $G$  is the divided-difference matrix

$$G_{ij} = \begin{cases} \frac{f_\epsilon(\lambda_i) - f_\epsilon(\lambda_j)}{\lambda_i - \lambda_j}, & i \neq j, \\ f'_\epsilon(\lambda_i), & i = j, \end{cases} \quad f_\epsilon(\lambda) = (\lambda + \epsilon)^{-1/p}.$$

This representation makes two facts explicit.

(i) *Diagonal drift can dominate even when the off-diagonal residual is tiny.* The diagonal entries  $(Q^\top \Delta X Q)_{ii}$  are weighted by

$$|f'_\epsilon(\lambda_i)| = \frac{1}{p}(\lambda_i + \epsilon)^{-(p+1)/p},$$

which is the largest near the smallest eigenvalues. Therefore, even when the stale basis nearly diagonalizes  $X$  in the usual residual sense, small diagonal drift in low-spectrum directions can induce a large change in the inverse-root operator. The counterexample above is exactly of this type.

(ii) *Off-diagonal mass is not uniformly important.* A diagonalization residual treats all off-diagonal entries of  $\hat{Q}^\top X \hat{Q}$  on roughly equal footing. In contrast, the inverse-root response depends on *where* those off-diagonal components occur through the divided differences  $G_{ij}$ . Mixing between well-conditioned high-eigenvalue directions may be relatively harmless, whereas perturbations that act on, or mix into, low-spectrum directions can be strongly amplified. Thus, the inverse-root map is intrinsically anisotropic, and a uniform off-diagonal residual is not aligned with the actual update sensitivity.

**Implication for decision rules.** The preceding discussion shows that the question

$$\text{“How diagonal does } \hat{Q}^\top L_t \hat{Q} \text{ look?”}$$

is not the same as

$$\text{“How accurate is } (L_t + \epsilon \mathbf{I}_m)^{-1/p} \text{ under the stale approximation induced by } \hat{Q} \text{?”}$$

The former concerns matrix approximation in a stale eigenspace; the latter concerns the fidelity of a nonlinear, spectrum-sensitive operator that directly enters the update. Accordingly, a decision rule based only on diagonalization residual may be poorly calibrated for Shampoo-style stability control.

For this reason, we do not use the diagonalization-residual trigger of Eschenhagen et al. (2026) as the principal control signal. Instead, we target the relative inverse-root operator error itself. Concretely, our sensing objective is the relative gap

$$\frac{\|\Delta_L(t)\|_F}{\|\hat{L}_t^{-1/p}\|_F}, \quad \Delta_L(t) := (L_t + \epsilon_t \mathbf{I}_m)^{-1/p} - \hat{L}_t^{-1/p},$$

and our proxy is derived from a first-order approximation to this quantity in the stale eigenspace. In particular, it reweights the drift by inverse spectral factors and therefore tracks precisely the low-spectrum sensitivity that the diagonalization residual misses.

| Model | Residual threshold ( $\tau$ ) | QR iters | Median runtime / direct EVD | Median residual after QR |
|-------|-------------------------------|----------|-----------------------------|--------------------------|
| ViT   | 0.1                           | 1        | $0.52\times$                | 0.695                    |
| ViT   | 0.1                           | 2        | $1.01\times$                | 0.451                    |
| ViT   | 0.1                           | 4        | $1.97\times$                | 0.230                    |
| ViT   | 0.1                           | 8        | $3.93\times$                | 0.098                    |
| ViT   | 0.25                          | 1        | $0.36\times$                | 0.682                    |
| ViT   | 0.25                          | 2        | $0.65\times$                | 0.360                    |
| ViT   | 0.25                          | 4        | $1.27\times$                | 0.125                    |
| ViT   | 0.25                          | 8        | $2.55\times$                | 0.060                    |

Table 3. QR refinement cost to satisfy the diagonalization-residual criterion on saved ViT trigger states.

**On the (mis)use of warm-started QR: a cold-started schedule in disguise.** The idea of refining a stale basis by a small number of QR iterations is natural. Our criticism is not of warm-started QR *per se*, but of the regime in which it is invoked under the trigger policy of Eschenhagen et al. (2026). Their QR refinement is activated precisely when the residual criterion fails, i.e. when the current basis is already a poor diagonalizer of the current statistic. In that regime, the iterate is no longer necessarily close to the desired invariant subspace, so the favorable local behavior associated with a genuine warm start cannot be assumed. Operationally, the refinement is therefore often initiated in what is better described as a *cold-started regime*.

This matters computationally. Dense QR iteration costs  $\mathcal{O}(n^3)$  per step (and similarly  $\mathcal{O}(m^3)$  for the other factor), and several iterations may be required before entering an accurate regime. In contrast, a direct eigensolver such as `eigh` performs the standard reduction once and then solves the resulting tridiagonal problem efficiently. Hence, when the trigger calls QR only after the basis has already drifted substantially, the cumulative cost of multiple dense QR iterations can be comparable to, or exceed, that of direct eigendecomposition.

This interpretation is also consistent with the empirical findings reported by Eschenhagen et al. (2026): on their Imagewoof ViT experiment, adaptive QR was slower in wall-clock time than adaptive `eigh`, and even the default one-step SOAP-style refinement was slightly slower and reached a worse final loss. Indeed, they ultimately emphasize using the criterion mainly to decide when to call `eigh`, rather than to justify QR refinement as the primary computational primitive. From our viewpoint, this is exactly what one would expect when the trigger is not aligned with the inverse-root operator error and when the refinement is invoked only after the stale basis has already become substantially inaccurate.

### I.1. Real-task complement: cost of residual-based QR refinement

We examine a complementary practical question on real trigger states collected during ViT training: how expensive is it to satisfy the diagonalization-residual criterion by QR refinement? Each saved state contains a stale basis together with the corresponding current Kronecker factor at a refresh-check step, and all runtimes are normalized by the direct EVD time measured on the same state and implementation path.

Table 3 reports the median number of QR iterations required to satisfy the residual threshold  $\tau$  in ? for these saved trigger states.

The median residual drops below  $\tau = 0.1$  (or  $\tau = 0.25$ ) only after 8 (or 4) QR iterations, at which point the median runtime is already  $3.93\times$  (or  $1.27\times$ ) that of direct EVD. Thus, for real saved ViT trigger states, a single QR step is typically insufficient to satisfy the residual criterion, whereas sufficient QR refinement to satisfy it is already slower than direct EVD.

Table 4. Hyperparameters

| (a) ViT on ImageNet. |                                             | (b) Conformer on LibriSpeech. |                                       |
|----------------------|---------------------------------------------|-------------------------------|---------------------------------------|
| Baselines            |                                             | Baselines                     |                                       |
| Hyperparameter       | Value                                       | Hyperparameter                | Value                                 |
| Epoch                | 90                                          | Epoch                         | 90                                    |
| Batch size           | 1024                                        | Batch size                    | 1024                                  |
| Momentum             | (0.95, 0.995)                               | Momentum                      | (0.95, 0.995)                         |
| WD                   | $4.2 \times 10^{-4}$                        | WD                            | $5.0 \times 10^{-4}$                  |
| LR                   | $\{1.75, 2.00, 2.20, 2.35\} \times 10^{-3}$ | LR                            | $\{1.75, 2.00, 2.25\} \times 10^{-3}$ |
| LR scheduling        | Warm-up (5%) + cosine decay                 | LR scheduling                 | Warm-up (5%) + cosine decay           |
| LR grafting          | Adam                                        | LR grafting                   | Adam                                  |

| (c) GPT-2 on WikiText. |                                    |
|------------------------|------------------------------------|
| Baselines              |                                    |
| Hyperparameter         | Value                              |
| Epoch                  | 30                                 |
| Batch size             | 512                                |
| Momentum               | (0.95, 0.99)                       |
| WD                     | $1.0 \times 10^{-2}$               |
| LR                     | $\{1.0, 3.5, 6.0\} \times 10^{-3}$ |
| LR scheduling          | Warm-up + cosine decay             |
| LR grafting            | Adam                               |

## J. Further Empirical Results

In this section, we discuss additional empirical results.

### J.1. Experimental Settings

We implement FOAM and all baselines in PyTorch 2.8 with CUDA 12.8. For the ViT (ImageNet-1k) task, we used the A6000  $\times 4$  units; as an exception, we used Pro6000  $\times 4$  units only for the experiment shown in Figure 4. Next, we used Pro 6000  $\times 4$  units for the Conformer (Librispeech) task, and we used B200  $\times 2$  units for the language model task. Unless stated otherwise, we use an identical batch size, momentum coefficients, step size, and its scheduling, and weight decay (WD) for AdamW, stale Shampoo (SS)<sup>3</sup>, FOAM, Eschenhagen et al. (2026), and SOAP<sup>4</sup> (Table 4a–4c). We couldn’t find the source code for Eschenhagen et al. (2026); hence, we implemented it.

For the learning rate (LR), we selected a single LR using only the SS setting (from the candidate grid in the tables) and then compared baseline optimizers and FOAM under this LR. We apply Adam LR grafting (Shi et al., 2023) for all experiments except AdamW and SOAP. Additional baselines and FOAM specific hyperparameters (*e.g.*,  $\mathbf{f}$ ,  $\tau$ ,  $\epsilon_{\max}$ ) are reported in each Figure.

### J.2. ViT on ImageNet: More Results

Our experimental results in Figure 6 show the validation accuracy and training loss as functions of wall-clock time (minutes). In general, FOAM exhibits a wall-clock advantage without loss of performance. This demonstrates that reducing the eigendecomposition and adjusting epsilon via the error proxy effectively controls both wall-clock time gains and staleness-induced preconditioner errors.

**Computational Overhead analysis** In table 5, we measured proxy-computation time and EVD time separately and reported each as a percentage of the total optimizer time over the full run. In table 6, we also recorded (Kronecker) factor-level timings by matrix dimensions, which showed that across ViT factor sizes (384-1024), proxy computation

<sup>3</sup><https://github.com/facebookresearch/optimizers.git>

<sup>4</sup><https://github.com/nikhilvyas/SOAP.git>



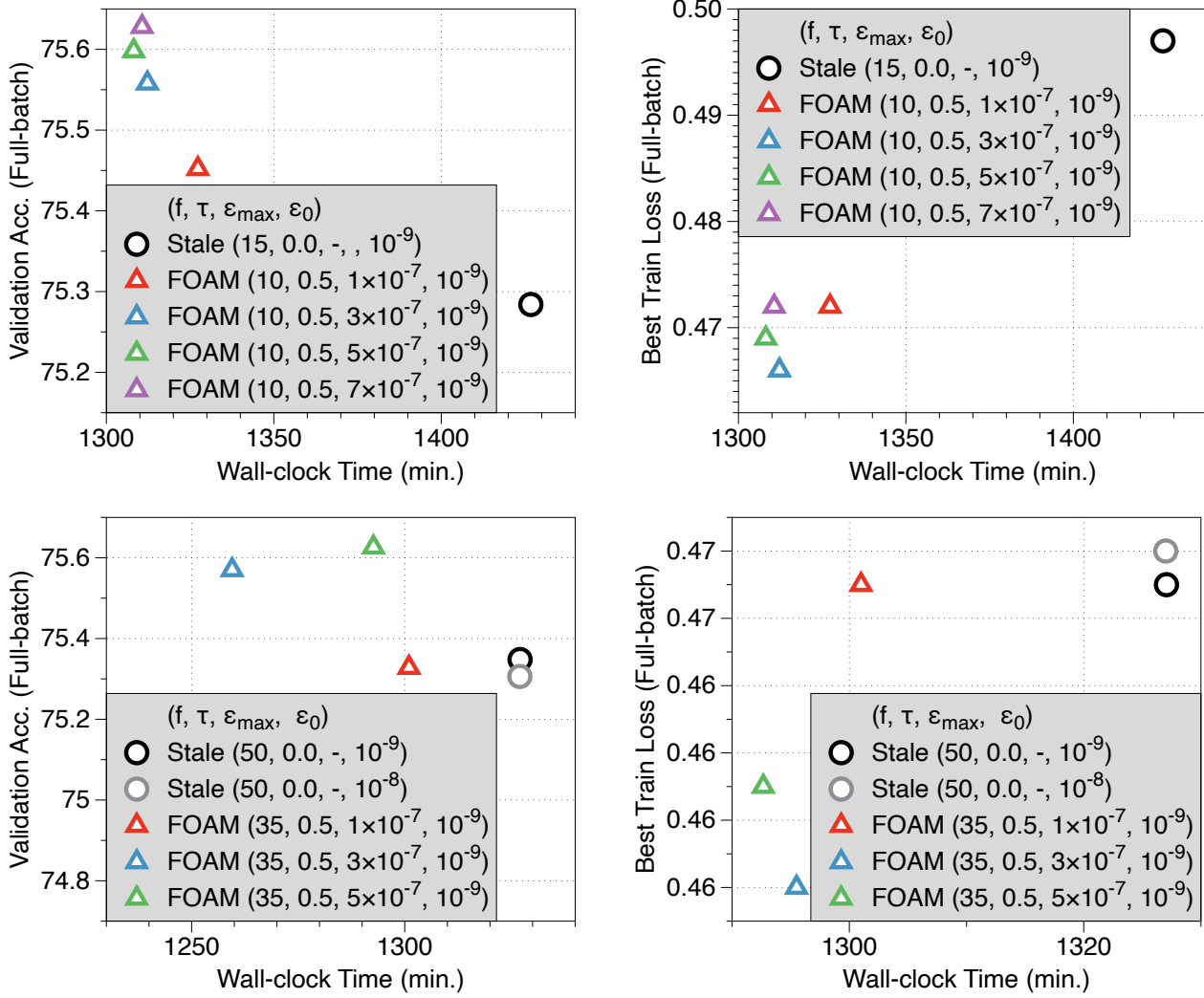


Figure 6. ViT: Best training loss and validation accuracy

remained nearly flat and was much smaller than the EVD cost. Thus, when comparing the corresponding median EVD times of 13.29 ms, 18.44 ms, and 14.98 ms, the proxy remained roughly  $50 - 70\times$  cheaper than EVD across the real-factor dimensions encountered during training.

### J.3. Conformer on LibriSpeech: More Results

Our experimental results in Figure 7 show the validation WER and training loss as functions of wall-clock time (minutes). In general, FOAM exhibits a wall-clock advantage without loss of performance. However, the FOAM  $(50, 0.3, 7 \times 10^{-7}, 10^{-9})$  setting exhibits a higher training loss than Stale Shampoo, indicating that excessive damping degrades the preconditioning effect.

### J.4. Language model on Wikitext-103-v1

We conducted experiments with the GPT-2 base (247M) model and confirmed FOAM’s robust effectiveness and superior efficiency (see Table 7).

| Model | Setting                                                             | Proxy computation cost | EVD cost       |
|-------|---------------------------------------------------------------------|------------------------|----------------|
| ViT   | $(\tau = 0.5; \mathbf{f} = 25; \epsilon_{\max} = 3 \times 10^{-7})$ | $\sim 4.72\%$          | $\sim 17.67\%$ |
| ViT   | $(\tau = 0.1; \mathbf{f} = 50; \epsilon_{\max} = 2 \times 10^{-7})$ | $\sim 2.10\%$          | $\sim 29.48\%$ |

Table 5. Proxy ( $h_t$ ) computation cost and EVD cost under different settings.

| Model & Setting                                                           | Dimensions | profiling |
|---------------------------------------------------------------------------|------------|-----------|
|                                                                           | 384        | 0.256 ms  |
| ViT ( $\tau = 0.5; \mathbf{f} = 25; \epsilon_{\max} = 3 \times 10^{-7}$ ) | 512        | 0.264 ms  |
|                                                                           | 1024       | 0.286 ms  |

Table 6. Dimension-wise profiling results for ViT.

4.

### J.5. Compare with Eschenhagen et al. (2026)

For the diagonal-residual criterion method (Eschenhagen et al., 2026), we refresh using EVD rather than QR iterations. Our reason is that, as observed in Table 3, the number of QR iterations needed to satisfy the residual criterion was already comparable to, or slower than, direct EVD. Furthermore, although Eschenhagen et al. (2026) proposed adaptive QR, they also observed in their experiments that the QR algorithm was less efficient than EVD in terms of wall-clock time and explicitly stated that they used EVD when the diagonal-residual was greater than the threshold  $\tau$  (refer to subsection 4.2 in Eschenhagen et al. (2026)). Taking these factors into account, we conducted experiments using EVD rather than warm-started QR iteration. We report additional experimental results in Table 8.

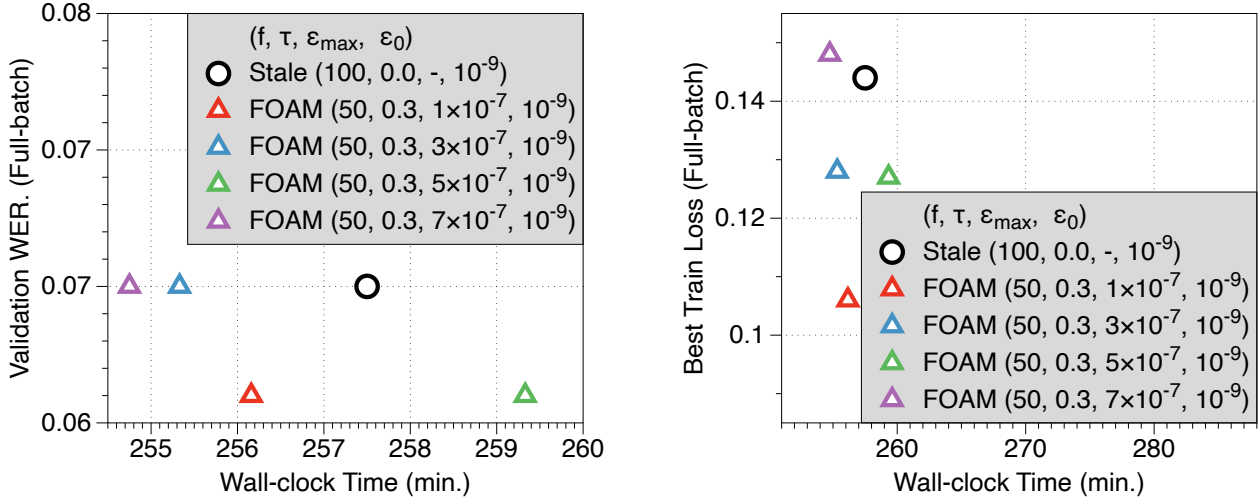


Figure 7. Conformer: Best training loss and validation WER

| Optimizer  | (f, $\tau$ , $\epsilon_0$ , $\epsilon_{\max}$ ) | Train Loss | Val PPL | Wall-clock Time |
|------------|-------------------------------------------------|------------|---------|-----------------|
| AdamW      | (N/A, N/A, $10^{-12}$ , N/A)                    | 2.30       | 18.37   | 609 (minute)    |
| Shampoo    | (100, N/A, $10^{-12}$ , N/A)                    | 2.31       | 18.26   | 630 (minute)    |
| FOAM       | (75, 0.4, $10^{-12}$ , $1 \times 10^{-10}$ )    | 2.29       | 18.22   | 622 (minute)    |
| FOAM       | (75, 0.4, $10^{-12}$ , $2 \times 10^{-10}$ )    | 2.30       | 18.22   | 621 (minute)    |
| FOAM       | (75, 0.4, $10^{-12}$ , $3 \times 10^{-10}$ )    | 2.30       | 18.27   | 625 (minute)    |
| DR-Shampoo | (75, 0.4, $10^{-12}$ , N/A)                     | 2.31       | 18.33   | 623 (minute)    |
| SOAP       | (75, N/A, $10^{-12}$ , N/A)                     | 2.31       | 18.33   | 642 (minute)    |

Table 7. Comparison of wall-clock time and training loss on the language model task.

| Optimizer  | (f, $\tau$ , $\epsilon_0$ , $\epsilon_{\max}$ ) | Train Loss | Validation accuracy | Wall-clock Time |
|------------|-------------------------------------------------|------------|---------------------|-----------------|
| DR-Shampoo | (20, 0.25, $10^{-9}$ , -)                       | 0.489      | 74.7%               | 1,354 (minute)  |
| FOAM       | (20, 0.25, $10^{-9}$ , $3 \times 10^{-7}$ )     | 0.468      | 75.38 %             | 1,310 (minute)  |
| DR-Shampoo | (20, 0.5, $10^{-9}$ , -)                        | 0.494      | 74.2 %              | 1,326 (minute)  |
| FOAM       | (20, 0.5, $10^{-9}$ , $3 \times 10^{-7}$ )      | 0.467      | 75.6%               | 1,300 (minute)  |
| DR-Shampoo | (20, 0.75, $10^{-9}$ , -)                       | 0.531      | 73.56 %             | 1,310 (minute)  |
| FOAM       | (20, 0.75, $10^{-9}$ , $3 \times 10^{-7}$ )     | 0.458      | 75.72 %             | 1,300 (minute)  |

Table 8. Comparison of wall-clock time and training loss on the ViT task with DR-Shampoo Eschenhagen et al. (2026) and Ours.

## K. Proxy Validation Experiments

### K.1. Synthetic example

**Goal.** We test whether the surrogate

$$h(\epsilon) = RC(\epsilon) \frac{\alpha(\epsilon)}{p}$$

tracks the ground-truth relative inverse-root operator error and yields a useful trigger signal for eigendecomposition.

**Data generation.** For each configuration, we sample an SPD matrix

$$A = Q \text{diag}(\lambda) Q^\top, \quad \lambda_i = i^{-r},$$

where  $Q$  is obtained by QR factorization of a Gaussian matrix. To model controlled stale drift, we use a PSD perturbation

$$E = BB^\top \succeq 0,$$

rescaled as

$$E \leftarrow \frac{E}{\|E\|_F} (\text{E\_scale} \cdot \|A\|_F), \quad A_{\text{new}} = A + E.$$

We use PSD perturbations to preserve positive definiteness uniformly across the sweep.

**Damping sweep and operators.** We sweep  $\epsilon$  on a logarithmic grid

$$\epsilon \in [\epsilon_{\min}, \epsilon_{\max}]$$

with  $K = \text{eps\_steps}$  points. For inverse-root order  $p$ , define the stale and fresh damped inverse-root operators by

$$P_s(\epsilon) = Q \text{diag}((\lambda + \epsilon)^{-1/p}) Q^\top, \quad P_f(\epsilon) = U \text{diag}((\tilde{\lambda} + \epsilon)^{-1/p}) U^\top,$$

where

$$(\tilde{\lambda}, U) = \text{EVD}(A_{\text{new}}).$$

**Ground-truth error.** We measure the target relative operator error by

$$\Delta(\epsilon) = \frac{\|P_f(\epsilon) - P_s(\epsilon)\|_F}{\|P_s(\epsilon)\|_F}.$$

**Proxy and baseline.** Let

$$\hat{E} = Q^\top E Q, \quad D = \text{diag}(\lambda + \epsilon).$$

We compute

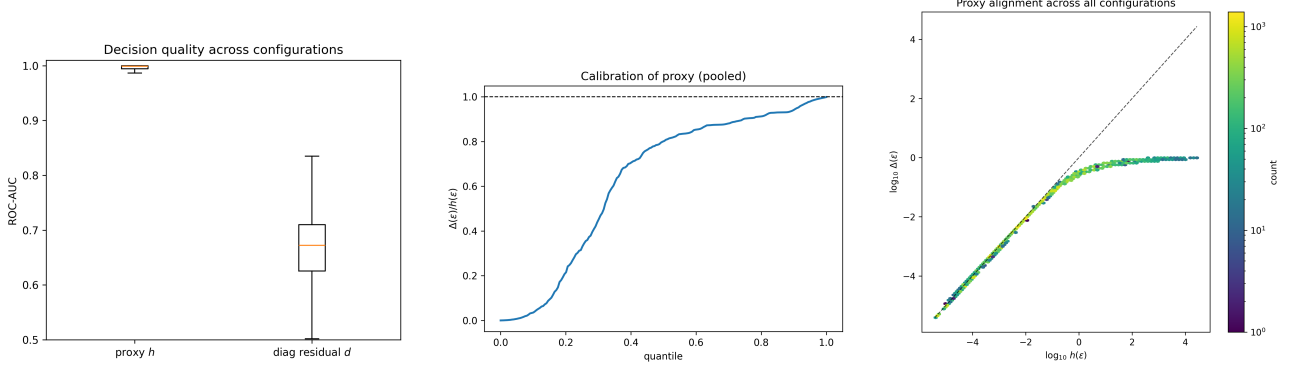
$$RC(\epsilon) = \|D^{-1/2} \hat{E} D^{-1/2}\|_F, \quad \alpha(\epsilon) = \frac{\|(\lambda + \epsilon)^{-1/p}\|_\infty}{\|(\lambda + \epsilon)^{-1/p}\|_2}, \quad h(\epsilon) = RC(\epsilon) \frac{\alpha(\epsilon)}{p}.$$

As a baseline, we use a *damped* diagonalization-residual score in the stale basis:

$$L(\epsilon) = U \text{diag}(\tilde{\lambda} + \epsilon) U^\top, \quad \hat{L}(\epsilon) = Q^\top L(\epsilon) Q, \quad d(\epsilon) = \frac{\|\hat{L}(\epsilon) - \text{diag}(\hat{L}(\epsilon))\|_F}{\|\hat{L}(\epsilon)\|_F}.$$

We include the same damping level  $\epsilon$  in  $d(\epsilon)$  so that the proxy and the baseline are compared under a matched damped operator scale.

**Evaluation.** For each configuration, we compute Pearson and Spearman correlations between  $\log_{10} \Delta(\epsilon)$  and  $\log_{10} h(\epsilon)$  across the  $K = 25$  values of  $\epsilon$ , and we compute the ROC-AUC for detecting “refresh-needed” samples defined as the top 20% by  $\Delta(\epsilon)$  within that configuration. We then summarize these configuration-wise statistics by their median and interquartile range (IQR) across all configurations. For calibration and alignment, we additionally pool all samples across all configurations and examine (i) the empirical distribution of  $\Delta(\epsilon)/h(\epsilon)$  and (ii) the log-scale scatter of  $\log_{10} h(\epsilon)$  versus  $\log_{10} \Delta(\epsilon)$ .



(a) **Decision utility (configuration-wise).** Distribution across the 150 configurations of ROC-AUC for identifying the top-20% samples by  $\Delta(\epsilon)$ . (b) **Calibration (pooled).** Empirical quantiles of  $\Delta(\epsilon)/h(\epsilon)$  over all 56,250 samples; values below 1 indicate conservativeness. (c) **Alignment (pooled).** Hexbin of  $\log_{10} h(\epsilon)$  versus  $\log_{10} \Delta(\epsilon)$  over all 56,250 samples, with  $y = x$  shown as a reference line.

**Figure 8. Synthetic validation of the proxy.** Panel (a) reports configuration-wise decision quality, showing that  $h(\epsilon)$  provides an almost perfect trigger signal for eigendecomposition and substantially outperforms the diagonalization-residual baseline  $d(\epsilon)$ . Panels (b) and (c) pool all samples across the sweep: the calibration plot shows that  $\Delta(\epsilon)/h(\epsilon) \leq 1$  throughout the evaluated range, indicating conservative behavior in this experiment, while the alignment plot shows strong monotone agreement between  $h(\epsilon)$  and the ground-truth relative operator error  $\Delta(\epsilon)$ .

**Sweep.** We sweep

$$d \in \{256, 512, 1024\}, \quad p \in \{2, 4\}, \quad r \in \{0.5, 1.0, 1.5, 2.0, 2.5\},$$

$$\text{E\_scale} \in \{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}, \quad \epsilon \in [10^{-8}, 10^{-2}]$$

(log-spaced), with  $K = 25$  damping values and  $T = 15$  trials per configuration. This yields 150 configurations and 56,250 total samples.

**Findings.** Across the 150 configurations, the proposed proxy  $h(\epsilon)$  provides near-perfect decision utility: the configuration-wise ROC-AUC has median 0.9988 (IQR [0.9944, 1.000]; worst-case 0.902), substantially outperforming the diagonalization-residual baseline  $d(\epsilon)$  (median 0.672, IQR [0.625, 0.710]). The configuration-wise log-scale alignment is also strong, with median Pearson correlation 0.9994 and median Spearman correlation 0.9982. For pooled calibration over all 56,250 samples, the ratio  $\Delta(\epsilon)/h(\epsilon)$  has median 0.761 and maximum 0.999, indicating that, throughout our sweep, the proxy remains conservative while staying tightly aligned with the target relative error; see Figure 8.